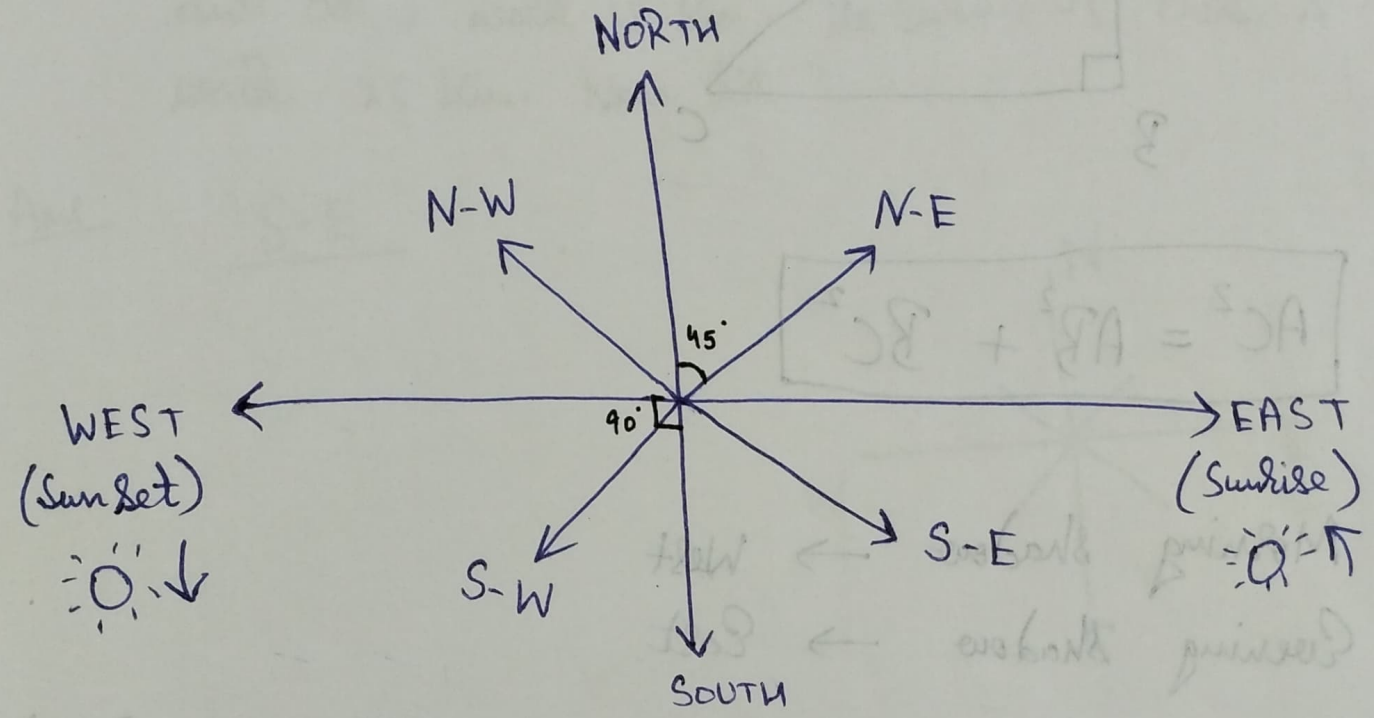
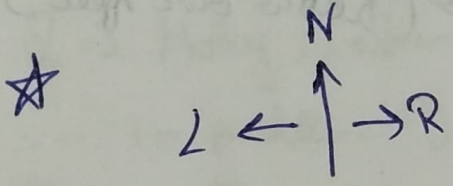
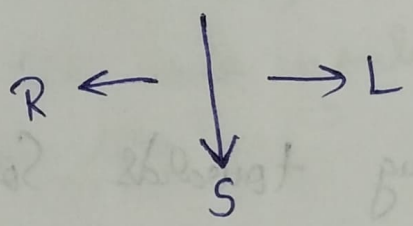
POORNIMA

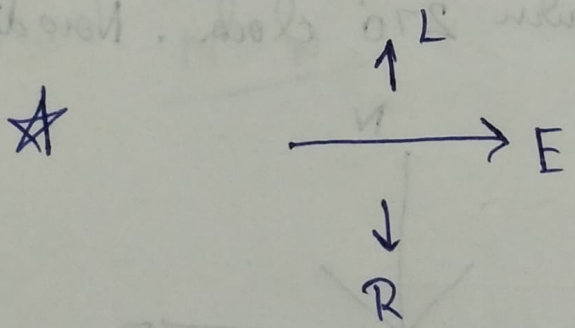
DIRECTIONS

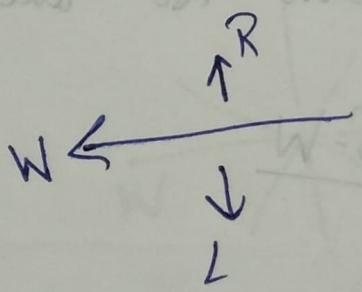
Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-





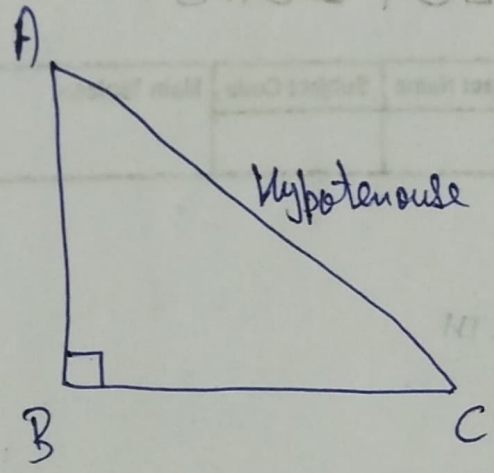






Main Ideas, Questions & Summary:

Library / Website Ref.:-



$$AC^2 = AB^2 + BC^2$$

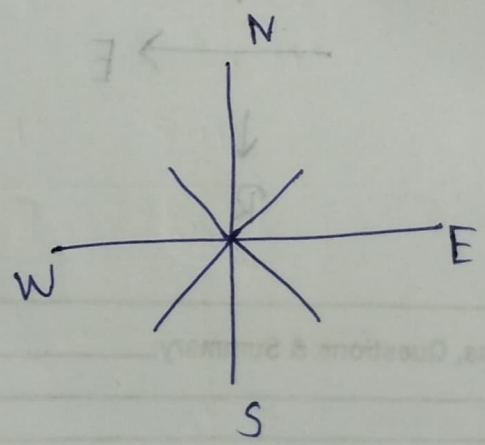
★ Morning shadow → West

Evening shadow → East

12 Noon shadow → No shadow (Below our feet)

Q ⇒ Joe facing towards South, turns 45° anti-clock. Turn 180° anti clock. Turn 270° clock. Now dir?

Ans S-W

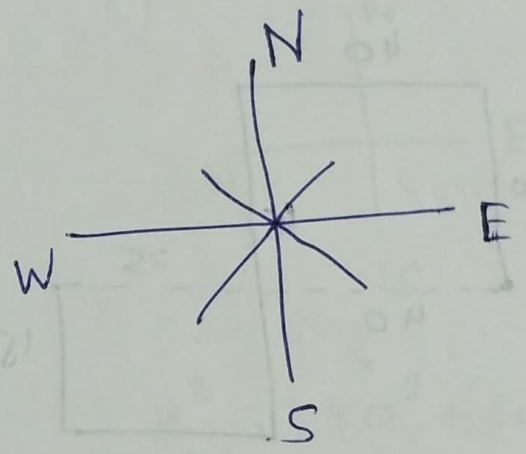


POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

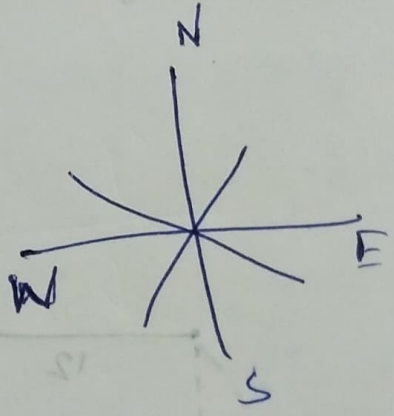
Q => Joe facing North, walk 10 km. Turn 270° anti clk, walk 15 km. He turn 45 clck & walk 25 km. Now dir?

Ans S-E



Q => Mary walking S-W 15 miles, Turn right, walk 10 miles. Turn 90° CLK walk 10 miles. Turn left walk 30 miles. Now direction?

Ans N-W



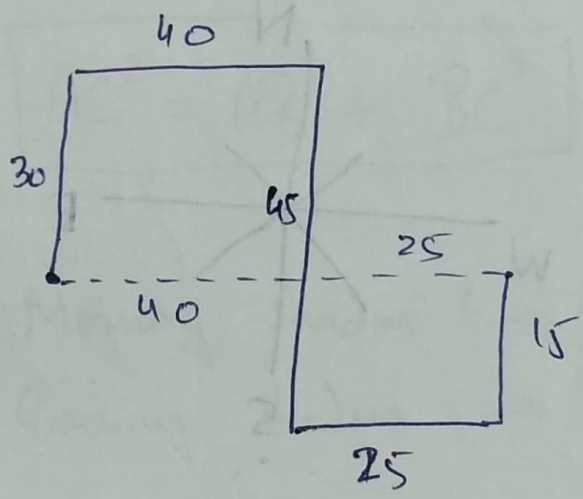
Main Ideas, Questions & Summary:

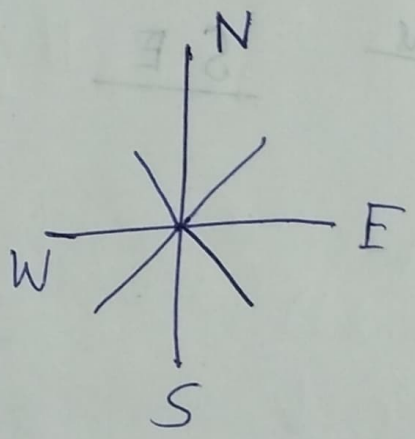
Library / Website Ref.:-

Q⇒ Maria walk 30 m North. Turn right walk 40 m. Turn right walk 45 m. Turn left walk 25 m. Turn left walk 15 m. How far from start position.

Ans

65 m

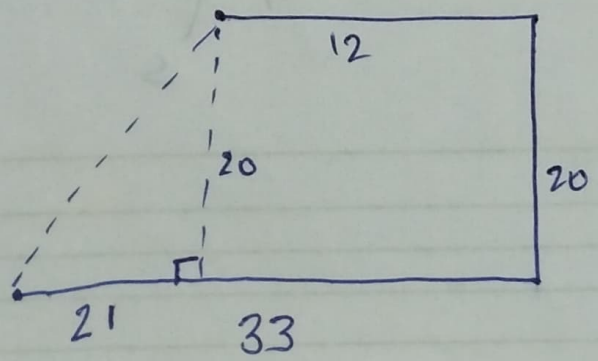




Q⇒ Boy walk 12 km east. Turn 40° clk walk 20 km. Turn right walk 33 km. Far from starting point?

Ans

$$H^2 = S_1^2 + S_2^2$$
$$= 20^2 + 21^2$$
$$H^2 = 841$$
$$\boxed{H = 29\text{ km}}$$

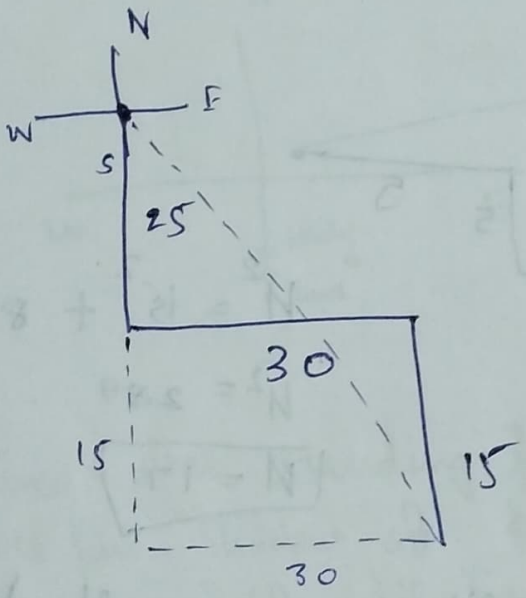


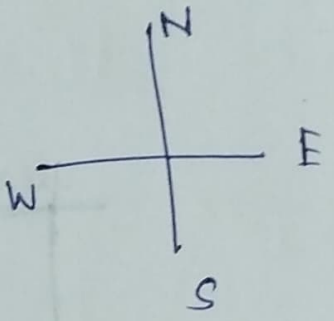
POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Car travel 25 Km South. Left turn travel 30 Km. turn right travel 15 Km. How far car is which dir?

Ans SE , 50 Km





$H^2 = 40^2 + 30^2$

$H^2 = 2500$

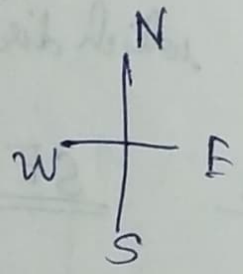
$H = 50$

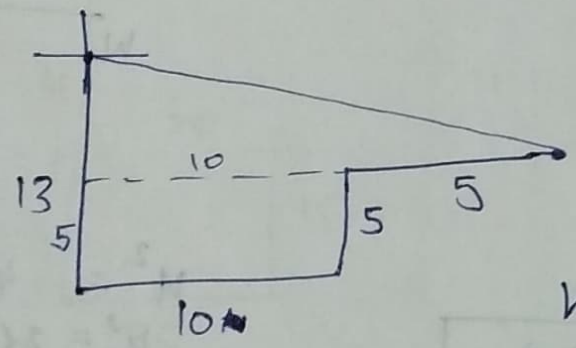
Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ Martin walked 13 m South from gym. Turned left walked 10m. Took left turn & walked 5 m. turn. Right. Walk for 5m. Far from gym & direction?

Ans SE , 17 m





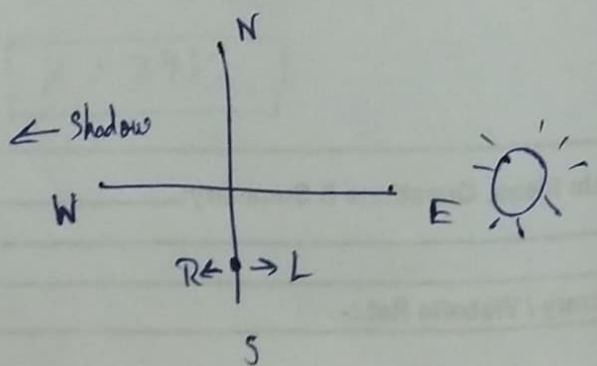
$H^2 = 15^2 + 8^2$

$H^2 = 289$

$H = 17$

Q⇒ One morning after sunrise, Joe was standing facing a pole. The shadow of pole fell exactly to his right. To which dir was he facing?

Ans Towards South

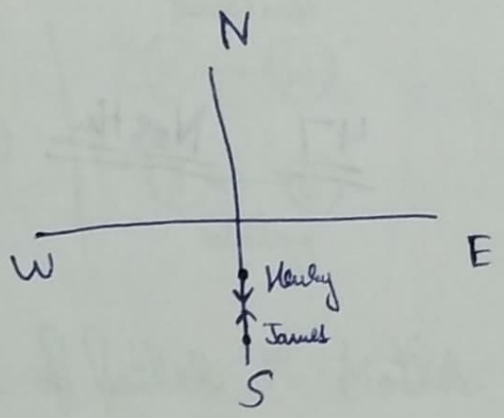


POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => James & Henery were standing facing each other at 8 am. Shadow of Henry fell exactly to his right. To which direction was James facing?

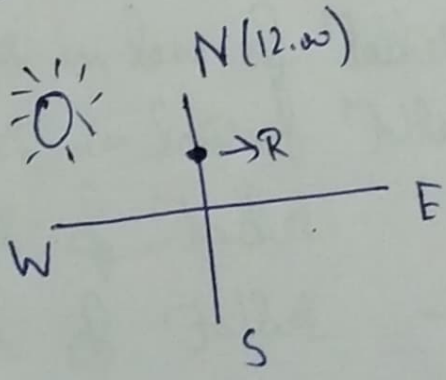
Ans



North

Q => Prince was standing facing to the pole at 1.48 pm. Shadow of pole fell towards his right. To which direction was Prince facing?

Ans



North

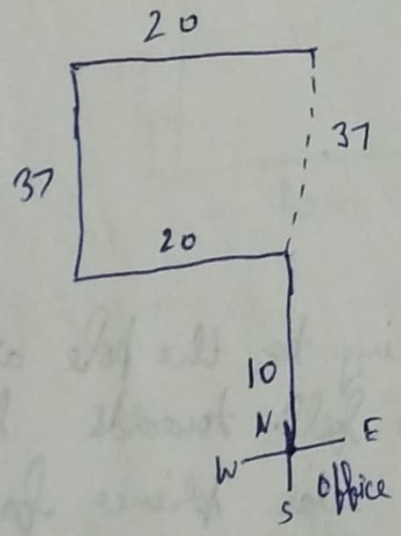
Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q => Ted drives a car 10 miles towards North from his office. He turns 270° clockwise & drives for 20 miles. Now, he turns 270° anti-clock. and drives a car for 37 miles. Finally, he reaches his house after driving for 20 miles to his right. How far is Ted's house from his office and in which direction.

Ans

47 North



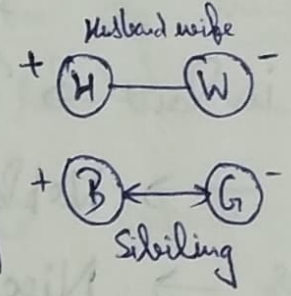
POORNIMA

BLOOD RELATIONS

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Draw diagram always for easy.
★ Every generation has new Level.
★ Use of symbols
★ Don't judge gender on basis of name.
★ Symbols ⇒

+ (Male)
- (Female)



★ Father of father or Mother → Grand Father
Mother of father or Mother → Grand Mother
Wife of Grandfather → Grand Mother
Husband of Grandmother → Grand Father
Father in law of Father/Mother → Grand Father
Mother in law of Father/Mother → Grand Mother
Brother of Father → Uncle
Sister of Father → Aunt

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ Brother / Sister of mother → Maternal uncle / Aunt

Father of wife / Husband → Father - in - law

Mother of wife / Husband → Mother - in - law

Father / Mother only Son's / daughter → oneself

Son / Daughter of Uncle / Aunt → Cousin

Brother / Sister of Husband or wife → Brother in law / Sister in law

Husband of sister / Sister in law → Brother in law

Son of Brother or Sister → Nephew

Daughter of Brother or sister → Niece

Husband of Daughter → Son in Law

Wife of brother / Brother in law → Sister in law

Wife of Son → Daughter in law

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Type 1 → Person introduce another person
→ Use back tracking

Q ⇒ Rupesh introduces Poonam as [daughter ③ of the only son ② of my father's wife ①]. How Poonam related to Rupesh?

Ans • Assume Rupesh as yourself

• ,

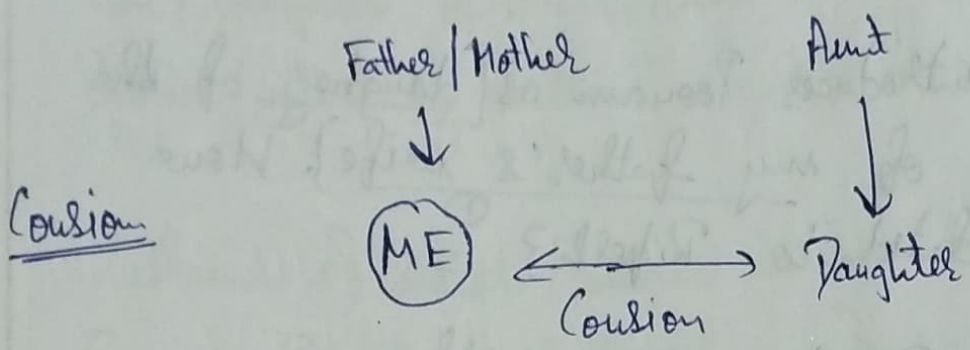
② only son
+
① my mother
↓
ME → Rupesh — daughter
↓
③ daughter

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Introducing a girl, a boy said "She is the daughter of mother of daughter of my aunt". How is girl related to boy?

Ans



Q ⇒ Type 2 ⇒ Symbols
⇒ Eliminate invalid option

Q ⇒ U + V means U is brother of V, W − X means W is father of S, X ÷ Y means X is sister of Y, Y × Z means Z is mother of Y. Which means N is mother of O?

(a) L + M ÷ N − O
(b) L − M × O ÷ P
(c) N ÷ M × L ÷ O
(d) M + L ÷ O × N

O × N

Ans (d)

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ If A # B means A is mother of B;
A - B means A is brother of B;
A % B means A is father of B;
A × B means A is sister of B, which of following P is maternal uncle of Q?

(a) Q - N # M × P
(b) P # S × N - Q
(c) P - M # N × Q
(d) Q - S % P

Option (C)

Ans

Mother - Brother (P)

(Q) (ME)

(a) +Q ↔ N
× |
M — P

(b) × P # S → P is female

(c) +P ↔ M
|
N ↔ Q

(d) Q ↔ S
× |
P

Main Ideas, Questions & Summary:

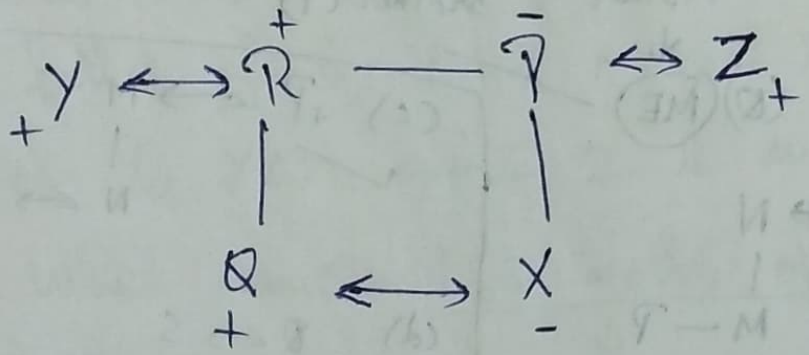
Library / Website Ref.:-

✡ Type 3 ⇒ Family tree

Q ⇒ ① A family consist of 6 people, P, Q, R, X, Y, Z
② Q is son of R but R is not mother of Q
③ P and R are married couple.
④ Y is brother of R, X is daughter of P.
⑤ Z is brother of P.

questions ⇒ (i) Who is brother in law of R?
(ii) How many female members are there.?
(iii) How is Q related to X?
(iv) How is Y related to P?

Ans



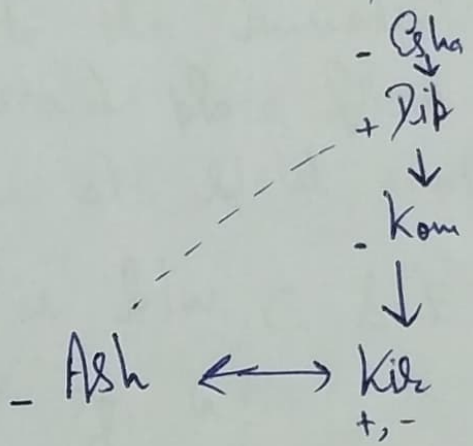
(i) Z
(ii) 2
(iii) Sibling, Brother
(iv) Brother in law of P

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Ashwini is Kiran's sister. Komal is Kiran's mother. Pipen is Komal's father. Esha is Pipen's mother. Then, how is Ashwini related to Pipen?

Ans



Grand daughter

Q ⇒ There are 6 persons Ashwin, Kiran, Komal, Pipen, Esha and Mittal. Komal is sister of Mittal. Kiran is brother of Esha's husband. Pipen is father of Ashwin & grandfather of Mittal. There are 2 fathers, 3 brothers & mother in group. Who is mother

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

SEAT ARRANGEMENT

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => A, B, C, D, E, F, G are 7 friends sitting in a single row facing North.

(1) D is to immediate right of C.

(2) E & A are neighbours of F.

(3) B is to the immediate left of C and on second place from left most end.

(4) A is at right most end.

(A) Who is b/w C & E?

(B) Neighbor of G?

(C) Position of E?

Ans Left Right

G B C D E F A

(A) D

(B) B

(C) 5 from left | 3 from right

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ ① 5 boys A, B, C, D, E & 5 girls K, P, R, S, V sit in a 2 rows facing towards each other. All the boys are in one row and all the girls in other row.

② E who is to the immediate right of B and opposite to P is not at any end.

③ R, who is immediate to right of K and opposite to C, at one of ends.

④ A is opp to K who is third to right of S.

⑤ D & V are not opposite each other.

Ans 5B 5G Right, Left carefully

D E B A C Boys ↓

S P V K R Girls ↑

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => 6 boys Ram, Su, Ri, Ru, Pa, Ka &
6 girls Se, Gr, Ri, Me, Ko, Rad are standing in row, such that each girl faces one boy, not necessarily in same order. [Seema is to the immediate right of girl who is facing Pa, the boy at extreme right.] [Only Su is b/w Ru & Pa.] [Kan is to immediate left of Ram & to immediate right of Ri.]
[Ri is facing Ram and it to the immediate left of Greeta. Radha is 3 to left of Me.

Ans 6B, 6G

Pa Su Ru Ram Kan ~~Pa~~ Ri Boys ↓

Ko Se Rad Ri Gr Me Girls ↑

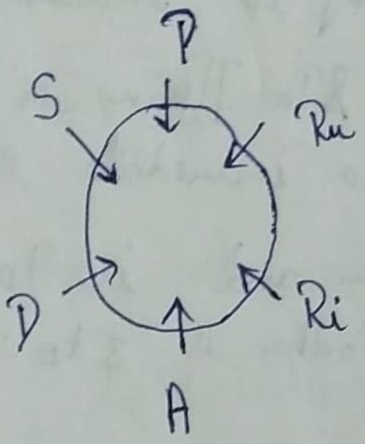
Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ 6 friends, P, Ru, A, D, Ri, S are sitting in a circle facing ~~towards~~ inwards

① P & A are opp to each other
② S is b/w P & D
③ D is immediately to left of A.
④ Ri is not exactly opp to Ri.

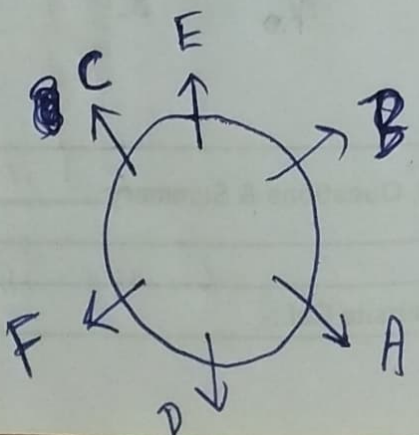
Ans



Q ⇒ 6 friends A, B, C, D, E, F are sitting on a circular table facing outwards.

① E is b/w B & C
② B & F are opp
③ A is exactly right of B.

Ans



POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ In a group A, B, C, D, E, F are sitting around a rectangular table.

① A is sitting on side alone

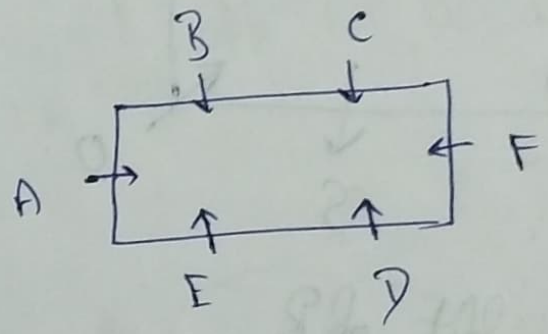
② B, C are sitting on same side

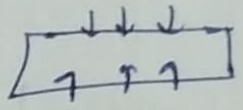
③ F is facing A

④ D is immediate left of F

⑤ C facing D.

Ans





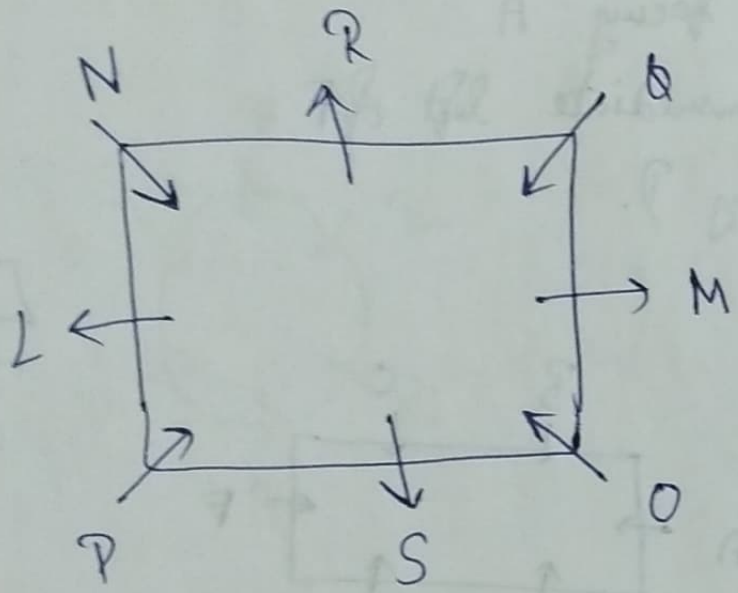
This Scenario can also be possible.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ L, M, N, O, P, Q, R, S are sitting around a sq table, such that 4 of them sit at 4 corner of sq, facing centre, while 4 sit in middle of each side, facing outward. 2 female sits in middle of sides and 2 at corners.

Ans



① L sits second to left of R.
② R sits in middle of one of sides
③ N sits fourth to right of his wife & his wife is not an immediate neighbour of L or R.
④ M sits thirds to right of her husband
⑤ M does not sit at any of corner
⑥ Only O sits b/w M and S
⑦ S is husband of L & P is male

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ ① Pe, Hu, Ir, Ha, Th, Lo, Bl are sitting in a circle.

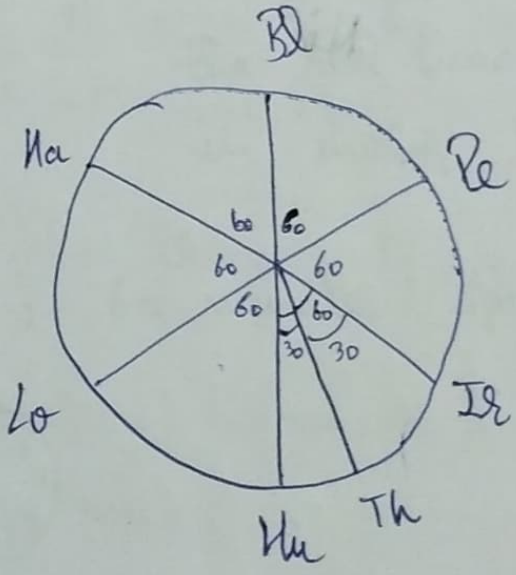
② Pe, Hu, Ir, Ha, Lo, Bl are sitting at equal distances from each other.

③ Ir is sitting 2 places right of Lo, who is sitting one place right of Ha.

④ Pe forms an angle of 90° from Th & an angle of 120° from Hu.

⑤ Hu is just opp to Bl and is sitting on left of Th

Ans



360/6 because ↓

6 are sitting at same distance

Main Ideas, Questions & Summary:

Library / Website Ref.:-

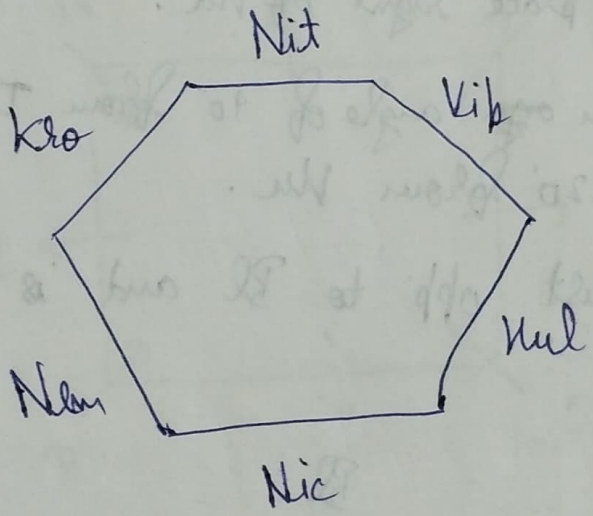
Q ⇒ ① 6 people Nit, Nem, Nic, Vip, Kro, Mul are attending a hexagonal table conference.

② All the sides of hexagon table are same length

③ Nit is not adjacent to Nem or Nic, Vip is not adjacent to Nic or Kro.

④ Nem & Nic are adjacent, Mul is in middle of Vip & Nic.

Ans



POORNIMA

CLOCKS

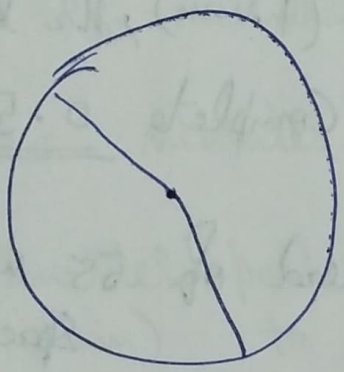
Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

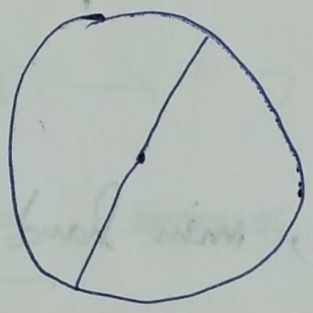
★ 24 Hrs 12 can be write as 00

1 Hr = 60 mins

1 min = 60 secs

★ Real | Mirror



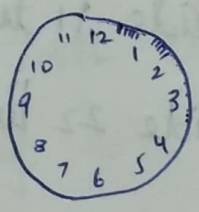


11 : 60
10 : 25
1 : 35

10 : 25

Always subtract from 11:60 the real time, to get time in mirror & Viceversa

★ In 12 Hr, 60 minutes spaces



Main Ideas, Questions & Summary:

Library / Website Ref.:-

✡ In 1 Hr, min hand move 360°
, Hr hand move 30°

✡ When Sec hand Complete 360°, min hand move 6°

✡ 1 Hr = 30°
1 min = 6°

✡ When min hand complete ~~360°~~ 6° (1 min), Hr hand Complete 0.5°

✡ For every 1 Hr, min hand is ahead of 55 min Spaces

✡ In 12 Hrs, we get 22 times 90° angle.
In 24 Hrs, " " 44 times 90° angle

✡ 3, Exception where only one 90° angle came.

✡ In 12 Hrs, min & Hr hand Coincide 11 times.
In 24 Hrs, " " " " Coincide 22 times.

12 is Exception, no coincide.

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ 12 Hrs, 180° (opposite each other) → 11 times

24 Hrs, " " " → 22 times

★ Straight line in 1 days → 22 + 22 → 44
(coincide) + (oppos)

★ 15 minutes past 7 → 7 : 15

★ 20 minutes to 8 → 7 : 40

★ Quarter past 6 → ~~5 : 45~~ 6 : 15
(15 min) to 6 → 5 : 45

★ Half past 6 → 6 : 30

★ Noon / midday → 12 : 00 PM

★ Midnight → 00 : 00

★ Broken clock is twice a per day.

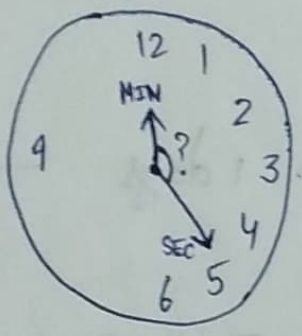
★ A fast/ loss clock is may or may not see twice per day.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ What angle is formed b/w the minutes hand and second's hand of clock if they are 25 min apart?

Ans



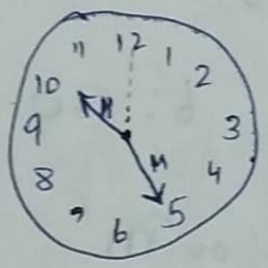
60 min space = 360°

1 min space = 6°

25 × 6° = 150°

Q ⇒ What will be angle b/w 2 hands of clock at 10:25 PM?

Ans



Hr ⇒

1 min space = 6°

50 × 6 = 300°

Min ⇒

25 × 6 = 150°

Wrong ⇒ 300° - 150°

⇒ 150°

* If minute hand on 25, hr hand will move slightly so for that

⇒ 25 × 0.5

⇒ 12.5°

⇒ 300° + 12.5 - 150°

⇒ 312.5 - 150

⇒ 162.5°

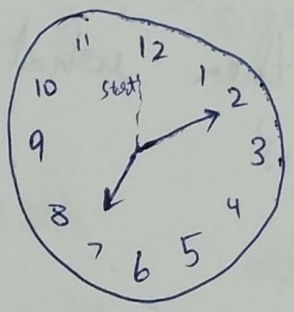
Ans

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => A clock is started at noon. By 10 min past 7, the hr hand has turned through ___

Ans



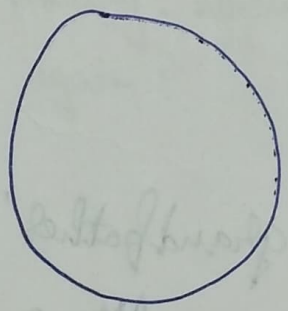
$35 \times 6^{\circ} = 210^{\circ}$

$10 \times 0.5^{\circ} = 5^{\circ}$

$210 + 5 \Rightarrow \underline{215^{\circ}}$

Q => If the min hand of clock has moved 300°, how many degrees has the hour hand moved?

Ans



1 min space = 6°

? = 300°

$\Rightarrow \frac{300^{\circ}}{6^{\circ}} = 50 \text{ min}$

$50 \times 0.5 = \underline{25^{\circ}}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q=> A clock when seen in a mirror shows 11:25. What is correct time?

Ans

$$\begin{array}{r} 11:60 \\ -\ 11:25 \\ \hline 0:35 \end{array} \longrightarrow 12:35$$

Q=> If real time 12:10, then what will be mirror image of that clock.?

Ans

$$\begin{array}{r} 11:60 \\ 12:10 \\ \hline \end{array}$$ 12 can also written as 00

$$\begin{array}{r} 11:60 \\ 00:10 \\ \hline 11:50 \end{array}$$

Q=> Ramesh sets his grandfather's clock to right time at 2:00 AM on thursday. But he didn't known that clock gains 20s for every 3 Hr. What will the clock indicate on Friday night 6:30 PM?

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Ans

2:00 AM → Thursday
2:00 AM → Friday } 24 Hr
2:00 PM → Friday } 12 Hr
6:30 PM → Friday } 4.30 Hr (4.5)

40.30 Hr (40.5)

20 Sec → 3 Hr
? → 40.5 Hr ⇒ 270 Sec ⇒ 4.5 min

Time will be 6:34:30

Q⇒ A clock loses 5 min every hr and was set right at 11 AM on Monday. When will it show correct time again?

Ans

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ My watch, which gains uniformly, is 2 minutes behind at noon on Sunday, and is 4 min 48 sec ahead at 2 PM on following Sunday. When was it correct?

Ans

Q ⇒ When will the clock's hands be in a straight line but pointing in opposite directions b/w 2 AM & 3 AM?

Ans

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ When will bee both hands of clock at right angle, b/w 8 PM & 9 PM?

Ans

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

CODING & DECODING

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

A	1
B	2
C	3
D	4
E	5
F	6
G	7
H	8
I	9
J	10
K	11
L	12
M	13
N	14
O	15
P	16
Q	17
R	18
S	19
T	20
U	21
V	22
W	23
X	24
Y	25
Z	26

For reverse position

27 - Original position

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ If SUNSHINE → TVOTIJOF

MOON ?

Ans NPPO

Q ⇒ WRONG → GNORW, RIGHT ?

↳ Connected in Ascending

Ans GHIRT

Q ⇒ SNOW → 7100 , WALL

Ans W A L L

23 + 1 + 12 + 12 ⇒ 4800

Q ⇒ CAP → 61 , PEN ?

Ans Reverse addition

P E N

16 5 14 (original)

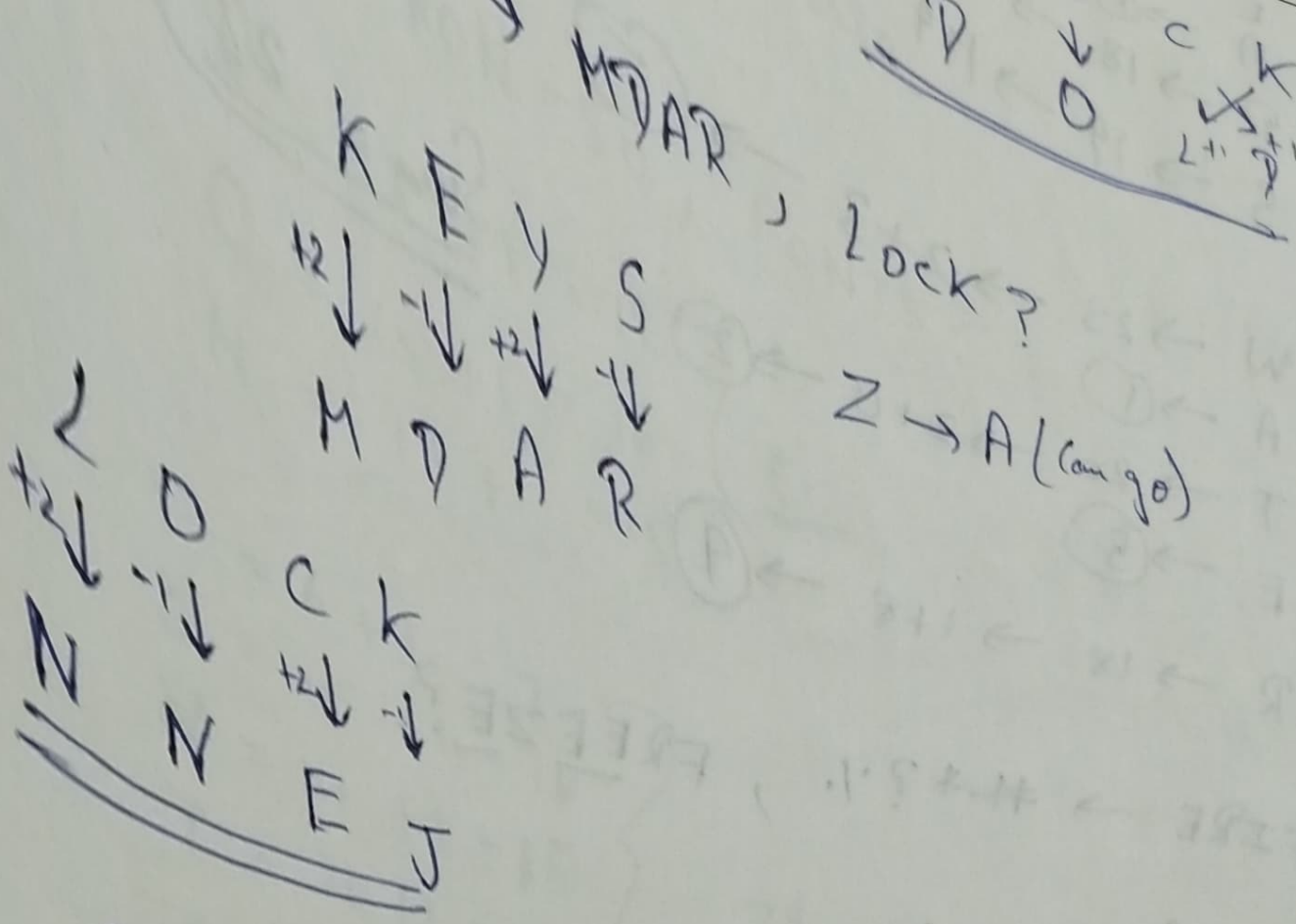
27-16 27-5 27-14 (For Reverse)

11 22 13 (Reverse)

46

MDAR, Lock?

Z → A (can go)



Q ⇒ RADAR → *?*?#, DOOR → ?*?*#, DAM?

Ans

R → *|#

A → *|#

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ TEARS → 18, WATER → ?

Ans T → 20 → 2+0 → (2)
E → (5)
A → (1)
R → 18 → 1+8 → (9)
S → 19 → 1+9 → 10 → 1+0 → (1)

Sum = 18

W → 23 → 2+3 → (5)
A → (1)
T → 20 → 2+0 → (2)
E → (5)
R → 18 → 1+8 → (9)

Sum = 22

Q⇒ FIRE → #*?%. , FREEZE?

Ans (a) #&%.%.?* X
(b) ??%.%.#* X
(c) %.%.%.?^^ X → Reapting so eliminate
(d) ???%.#@ ✓

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ THUMB → BMAHT, CRUMB?

Ans

T H (U) M B

B M (A) H T ← Reverse

↓ different

Q ⇒ R = 19, RON = 50, FONTS?

Ans

R = 18 + 1 = 19
O = 15 + 1 = 16
N = 14 + 1 = 15
} 50

F = 6 + 1 = 7
O = 15 + 1 = 16
N = 14 + 1 = 15
T = 20 + 1 = 21
S = 19 + 1 = 20
} 79

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ ACID → 1C3D

PAMPER → P1MP2R

BOMBAY → ?

Ans

A	E	I	O	U
1	2	3	4	5

B4MB1Y

Q ⇒ Blue → Green, Green → Yellow, Yellow → Orange, Orange → Black, Black → White, White → Red, Red → Pink, Pink → Brown, Brown → grey.

Color of human blood?

Ans

Red X

Pink X

Brown X

Grey X

White ✓ (Because White means Red)

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ dog → lion, lion → bison, bison → snake, snake → mongoose, mongoose → crocodile, which one is reared as pet?

Ans

Lion ✓

Bison ✗

Snake ✗

Crocodile ✗

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

STATEMENT & CONCLUSION

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

☆ Do not assume.
☆ Conclusion (Inference also)

Q ⇒ Statement : Teacher scolded Ram in front of all students.

Conclusion : The teacher disliked Ram

Ans False, Here conclusion is assumed, Maybe Ram was late to class so teacher scold.

Q ⇒ Statement : With the increase in consumption of petrol products, it is feared that petrol will be a rare commodity in near future.

Conclusion : Petrol product should be used efficiently.

Ans True

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Statement : Many people have been admitted in hospital. It is assumed that junk food is poisonous.

Conclusion : Junk food is always poisonous.

Ans False

Q ⇒ Statement ; Shyam is one of the probable student for securing 1st rank in class.

Conclusion : ① Shyam will secure 1st rank
② Shyam will not secure 1st rank.

Ans Either ① or ② follows.

Because Shyam may be secure 1st or may not.

Q ⇒ Statement : Due to recession, Company X has fired its 200 employees.

Conclusion : ① Company X is well known for firing people.
② All other competitors of company X are also affected because of recession.

Ans Neither ① & ② because ② is not related to statement.

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => Statement : I known nothing except the fact of my iguorance.

Conclusion : ① Writer is poor reader. He should read more to improve his knowledge.

② World of knowledge is a deep ocean to be explored & conquered by a single person.

Ans Only ② because ② is giving advice which will considerd as conclusion

Q => Statement : High IQ is important to be a scientist

Conclusion : ① Average student should not aspire to be a scientist.

② All highly gifted students become scientist

Ans Only ① because Moral / Ethics should not be considerd as ~~statement~~ conclusion.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Statement : If both the partners work as a team, marriage will be successful.
The couple celebrated golden jubilee of their wedding last year.

golden jubilee
↓
50 yr wedding

Conclusion : ① Both the partners are compatible with each other
② Some marriage do not last even 10 years nowdays.

Ans only ①

POORNIMA

CALENDAR

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Normal year ⇒ 12 months
↓
52 weeks
↓
365 days

365 | 7 → 52.1 → 52 weeks + 1 day (odd day)

↳ 364 days + 1 day (odd day)

★ If 1 Jan (Any day) → Wednesday, 31 Dec ? will be same as 1 Jan i.e. Wednesday.

★★ Begin & ends on same day of week.

★ 1 Jan → Wed, 31 Dec?

Wed --- Tue .... Wed --- Tue + 1 day
↳ Wednesday Ans

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ Leap year ⇒ Feb have 29 days

366 days
↓
52 weeks

★ 366/7 → 52.1 → 52 weeks + 2 days (odd days)

★ 1 Jan → Wed, 31 Dec

Wed -- Tue --- Wed --- Tue + 2 days
↳ Thursday Ans

★ To check Leap year ⇒

- Divide by 4 == 0 (Leap year) for Normal years (only last 2)
- Divide by 400 == 0 (Leap year) for century years (whole no)

- 1756 ✓
  1832 ✓
  1929 ✗

1600	✓
1700	✗
1800	✗
1900	✗
2000	✓
2100	✗

Series for century

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Does a leap year come after every 4 years?

Ans No, 1896 → 1900 → 1904
✓ X ✓

★ How many leap years in 100 years?

No. of Year	Leap year	Normal year
100	24	76
200	48	152
300	72	152
400	97	303

- (1 - 100) ⇒ 100 | 4 = 25 (LY)
  Remaining (100 - 25) = 75 (NY)
  Since 100 is not leap year, 25 - 1 ⇒ 24 (LY)
  75 + 1 ⇒ 76 (NY)

- For 400, 24 × 4 = 96
  76 × 4 = 304
  Since 400 is leap year,
  96 + 1 ⇒ 97
  304 - 1 ⇒ 303

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ Odd days ⇒

No. of odd days	Days of the week
0	Sunday
1	Mon
2	Tue
3	Wed
4	Thu
5	Fri
6	Sat

★ 10 days → Week?

$$7 \overline{)10} \quad (1, \text{ remainder } 3)$$

⇒ 1 week + 3 days

★ No of odds days in century ⇒

No's	LY	NY	No odd
100	24	76	$(24 \times 2) + (76) = 124 \mid 7 = 5$
200	48	152	$(48 \times 2) + (152) = 248 \mid 7 = 3$
300	72	228	1
400	97	303	0
500			5
600			3 (Pattern repeat)

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ What was day on 6 April 1896?

Ans ① Check odd days for 1895.

1800 + 95

↓ ↓

(3) 95/4

23 72

× ×

2 1

46 + 72 ⇒ (18)

⇒ $\frac{121}{7}$ ⇒ 17 W + (2 days)

② Jan'96 + Feb'96 + Mar'96 + 6 days April'96

↓ ↓ ↓ ↓

31 29 (1896 [Y]) 31 6

⇒ $\frac{97}{7}$ = 13 W + (6 days)

6 + 2 ⇒ 7

⇒ (1)

Monday (By odd Table)

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Day on 25 June 1983?

Ans ① odds days in 1982

⇒ 1900 + 82

↓ ↓ 82/4

① 20 62

(LY) (NY)

(20×2) + (62)

(102)

⇒ (103) ⇒ 14 W + (5 days)

② 31 + 28 + 31 + 30 + 31 + 25

176 ⇒ 25 W + (1 days)

⇒ 5 + 1 ⇒ 6

Saturday

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => Jan 1, 2001 → Monday, Dec 31, 2002 ?

Ans Jan 1, 2001 → Monday

31 Dec 2001 → Monday
(NY)

1 Jan 2002 → Tuesday

31 Dec 2002 → Tuesday
(N.Y)

Q => Calendar for 2007 will be same as calendar for nearest year in future.

Ans

2007	8	9	10	11	12	13	14	15	16	17	18
1	2	1	1	1	2	1	1	1	2	1	1

14 days

If no of odd days = 0 (Calender repeats after it)

$\frac{14}{7} = 0$

2018 (Calender repeat)

If 2018 is ~~then 2004~~ not a leap

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ Which day can't be last day of century?

Ans Tuesday, Thursday, Saturday

By 2 tables

Q⇒ How many days are there in "y" weeks, "y" days.

Ans

1 W = 7 days

y W = 7y days

y days = + y days

8y days

Q⇒ 15 Jan 1977 → Saturday. 15 Jan 1976?

Ans 15 Jan 1977 (saturday)

⑥

15 Jan 1976

~~leap~~ year

(2)

6-2 ⇒ ④

Thursday

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Calendar for month of April looks same as ?(maybe wrong) that for month of - - -

Ans

April	May	June	July
30	31	30	

30 + 31 + 30 ⇒ $\frac{91}{7} = 0$

July

Q ⇒ 11 April 1911 → Tuesday

** 17 Sept 1915 → ?

Ans 11 April 1911 → Tuesday (2) By table

11 April 1912 (LP) (2)

1913 → (1)

1914 → (1)

1915 → (1)

2 (Tuesday) + 5 ⇒ Sunday [11 April 1915]

19 + 31 + 30 + 31 + 31 + 17 ⇒ (159) / 7 ⇒ (5)

Sunday + 5 ⇒ Friday

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

SERIES

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Numeric Series ⇒

Q ⇒ 19 25 32 40 ? 59

+6 +7 +8 +9 +10

Ans 40 + 9 ⇒ 49

Q ⇒ 102 85 68 ? 34

-17 -17 -17 -17

Ans 68 - 17 ⇒ 51

Q ⇒ 1 2 6 24 120 ?

1×2 2×3 6×4 24×5 120×6

Ans 720

Q ⇒ 216 72 36 12 ? 2

$\frac{216}{3}$ $\frac{72}{2}$ $\frac{36}{3}$ $\frac{12}{2}$ $\frac{?}{3}$

Ans $\frac{12}{2} = 6$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

✯ Special Numeric Series ⇒

Q⇒ 0 3 8 15 24 35 ? 63

1^2-1 2^2-1 3^2-1 4^2-1 5^2-1 6^2-1 7^2-1 8^2-1

Ans

7^2-1 ⇒ 48

✯ Alphabet Series ⇒

Q⇒ 2E2 1H4 1K6 1N3 ?

Ans Reverse alphabet

22 19 16 13 10

-3 -3 -3 -3

1Q0

Q⇒ H_8 L_{12} O_{15} S_{19}

+4 +3 +4 +3

Ans V_{22}

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ 2, 5, 10, 17, 28, ?

Ans Prime difference

2 5 10 17 28 ?

3 5 7 11 13

28 + 13 ⇒ 41

Q⇒ 21 34 55 89 144 ? 377

13 21 34 55 89

Ans Fibonacci series

144 + 89 ⇒ 233

Q⇒ L O J Q H ?

Ans S

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ Find wrong term in following

10 14 28 34 64 68 136

4 14 6 30 4 68

Ans 34

4 difference
term addition - - -

Q⇒ Wrong

15 16 20 30 45 70

1 4 10 15 25

Ans 1^2 2^2 3^2 4^2 5^2

30

Q⇒ Wrong

1 3 10 21 63 129

2 7 11 42 66

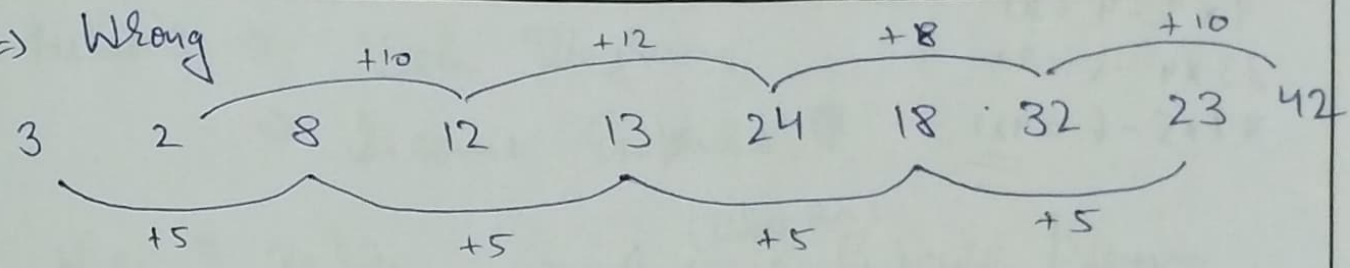
$(1\times2+1)$ $(3\times3+1)$ $(10\times2+1)$ $(21\times3+1)$ $(63\times2+1)$

Ans 63 It should be 62

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q => Wrong



Ans 24 (Should be 22)

Q => 1 11 34 75 139 ? (231)

10 23 41 64 92

Ans

13 18 23 28

5 5 5

Triangular Pattern

Q => 28 33 31 ? 34 39

+5 -2 +5 -2 +5

Ans

31+5 => 36

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q) 8, 6, 9, 23, 87 — ?

Ans

$8 \times 1 - 2 = 6$

$6 \times 2 - 3 = 9$

$9 \times 3 - 4 = 23$

$23 \times 4 - 5 = 87$

$87 \times 5 - 6 = 429$

POORNIMA

SYLLOGISM

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

☆ 2 Methods ⇒ Veen Diagram
⇒ Income Expense

☆ Normal Problem → Income Expense (Prefer this) / Venn
Possibility Problem → Venn diagram

Q ⇒ Statement : ① Some bottles (E_1(50)) are glass (E_2(50))
(Income) ② All the glasses (E_1(100)) are cables (E_2(50))

Conclusion : ① No glass (E_1(100)) are cable (E_2(100))
(Expense) ② Some cables (E_1(50)) are bottles (E_2(50))

Ans

	E_1	E_2
☆☆ All	100	50
No/None	100	100
Some	50	50
Some -- not	50	100

$I \geq E$

① Check conclusion ① which have same element in statement.
Stat E_2 > Concl E_2 (Cables), So conclusion ① Wrong X.

② A=B, B=C, A=C use in conclusion ②
If, at least one of the common element have 100, then we check.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

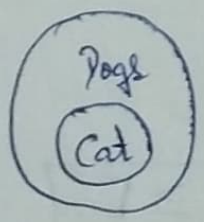
Q ⇒ Statement : ① No girl (100) is boy (100)
② All boys (100) are red (50)

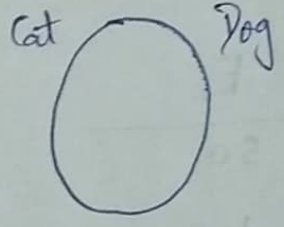
Conclusion : X ① No girl (100) is red (100)
✓ ② Some red (50) are boys (50)

Ans I ≥ E (Conclusion ②)

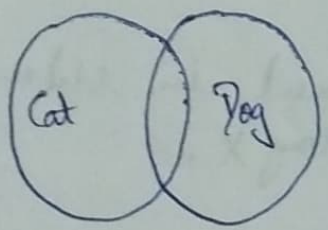
☆ Venn Diagram ⇒

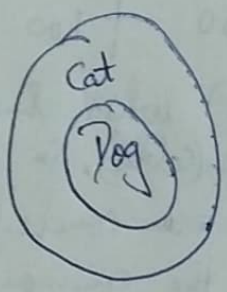
• All cats are dogs ⇒





• Some cats are dogs ⇒





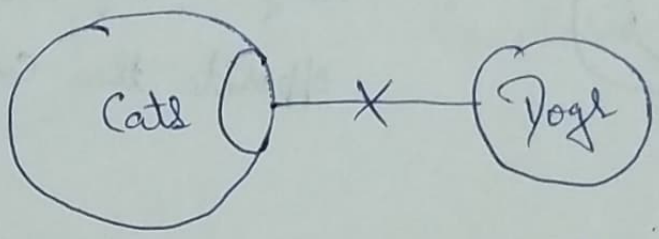
POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

- No cats are dogs ⇒



- Some cats are not dogs ⇒



Q⇒ Statement : ① Some cats are white.
② All white are dog
③ No dog is snow

Conclusion : ① No cats are dogs
② Some cats are dogs
③ No white is snow

Ans

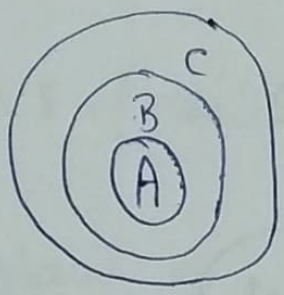
Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Statement : ① All A are B
② All B are C

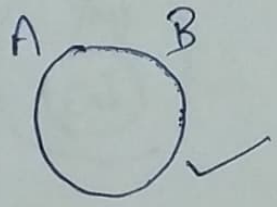
Conclusion : ① All B can be A
② Some C not being B is a possibility
③ Some C can be B.
④ Some A can be C

Ans



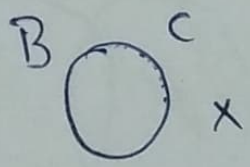
Draw another venn diagram, if answer is opposite then Conclusion true.

① Above diagram X



Since from one dig X & another ✓ i.e. Conclusion true.

② Above diagram ✓

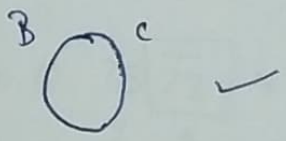


Conclusion true

POORNIMA

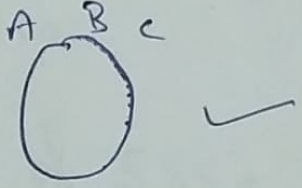
Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

(3) Above ✓



✓ Conclusion False

(4) Above ✓



✓ Conclusion False

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

FRACTIONS

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ What will be fractional form of $0.35\overline{23}$?

Ans ⇒ $0.35\overline{23} \rightarrow 0.35\,232323\ldots$

$2^{(0)}$ $2^{(9)}$

⇒

$$\frac{3523-35}{9900} \Rightarrow \frac{3488}{9900}$$

Q ⇒ Value of $5.55 + 5 + 5.555 + 5.05 + 5.00$

Ans

$$\begin{array}{r} 5.550 \\ 5.000 \\ 5.555 \\ 5.050 \\ +5.000 \\ \hline 26.155 \end{array}$$

} Make all 3 digit

Q ⇒ Value of $\dfrac{2.4 \times 0.72 \times 4.5}{0.18 \times 0.06 \times 0.9} \times 3$

Ans $\dfrac{\frac{24}{10} \times \frac{72}{100} \times \frac{45}{10}}{\frac{18}{100} \times \frac{6}{100} \times \frac{9}{10}} \times 3 \Rightarrow 2400$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Value of $\frac{m-m}{m+m}$ if 2.5 m = 0.035 m ?

Ans $\frac{25}{10}\ m = \frac{35\ m}{1000}$

$2500\ m = 35\ m$

$$\boxed{\frac{m}{m} = \frac{500}{7}}$$

$\frac{500-7}{500+7} \Rightarrow \frac{493}{507}$

Q ⇒ Add $17.4\overline{99}$, 17.85, $17.\overline{333}$?

Ans

$\Rightarrow 17 + 0.4\overline{99}$

$\Rightarrow 17 + \frac{499-4}{990}$

$\Rightarrow 17 + \frac{495}{990}$

$\Rightarrow 17.85$

$\Rightarrow 17 + 0.\overline{333}$

$\Rightarrow 17 + \frac{333}{999}$

$\Rightarrow 17.5$

17.85

Q ⇒ Value of $1 \times 0.3 \times 0.01 \times 0.003$

Ans $1 + 2 + 3$

6 decimals places

$1 \times 3 \times 1 \times 3$

$\Rightarrow \boxed{9} \Rightarrow 0.\underline{0}\,\underline{0}\,\underline{0}\,\underline{0}\,\underline{0}\,\underline{9}$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ Value of $\frac{1}{0.0004659}$ if $\frac{1}{4.659} = .2146$?

Ans $\frac{1}{0.0004659} \Rightarrow \frac{1000}{4.659} \Rightarrow 0.2146 \times 1000$

$\Rightarrow 2146$

Q⇒ Largest among $\frac{5}{8}, \frac{3}{7}, \frac{2}{9}, \frac{4}{5}$?

Ans ⇒ $\frac{5}{8} \times \frac{3}{7}$ | $\frac{2}{9} \times \frac{4}{5}$

$35 > 24$ | $10 < 36$

$\frac{5}{8}$ | $\frac{4}{5}$

$\frac{5}{8} \times \frac{4}{5}$

$25 < 32$

$\frac{4}{5}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Least among 0.2, $(.2)^2$, $0.\overline{2}$, $1 \div 0.2$

Ans 0.2

$(.2)^2 \Rightarrow (.04)$

$0.\overline{2} \Rightarrow \frac{2}{9} \Rightarrow 0.22$

$\frac{1}{0.2} \Rightarrow \frac{10}{2} \Rightarrow 5$

Q ⇒ Increasing Order $\frac{6}{7}, \frac{8}{9}, \frac{7}{8}, \frac{9}{10}$?

Ans $\mathcal{D} > N$ (all)

$\mathcal{D} - N$ = Same (all)

⇒ Small N → Small fraction
Greater N → Greater fraction

⇒ $\frac{6}{7} < \frac{7}{8} < \frac{8}{9} < \frac{9}{10}$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Increasing Order $\frac{14}{17}, \frac{10}{12}, \frac{6}{7}, \frac{18}{22}$

Ans $D > N$

But $D - N$ = Different

∴ Arrange N in increasing order

$\boxed{\frac{6}{7}} \xrightarrow{+4,+5} \boxed{\frac{10}{12}} \xrightarrow{+4,+5} \boxed{\frac{14}{17}} \xrightarrow{+4,+5} \boxed{\frac{18}{22}}$

Since increasing by is same in N & D

∴ Small N → Small fraction
Greater N → Greater fraction

Q ⇒ $\frac{4}{7}$ of $\frac{2}{3}$ of $\frac{5}{6}$ of $\frac{5}{8}$ of 1008 ?

Ans ⇒ $\frac{4}{7} \times \frac{2}{3} \times \frac{5}{6} \times \frac{5}{8} \times 1008$

⇒ 200

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ If a fraction denominator is decreased by 80% and numerator is increased by 300%. the fraction become 2/9. What is fraction?

Ans. Let $a \rightarrow N$

$b \rightarrow D$

⇒ a + 300% of a

$a + \frac{300}{100} \times a$

$4a$

⇒ b - 80% of b

$b - \frac{80}{100} \times b$

$0.2b$

$$\frac{4a}{0.2b} = \frac{2}{9}$$

$$\boxed{\frac{a}{b} = \frac{1}{90}}$$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ In 3 fractions when the largest fraction is divided by smallest fraction, the result 5/4 is greater than the middle fraction by 1/2. If sum of three fractions is $2\frac{3}{12}$ what will be the difference b/w largest & smallest fraction?

Ans Let Largest = A
Middle = B
Smallst = C

$\because \frac{A}{C} = \frac{5}{4}$ — (1)

$\therefore \frac{5}{4} = B + \frac{1}{2}$ — (2)

$\boxed{B = \frac{3}{4}}$

$\because A + B + C = \frac{27}{12}$

$\frac{5}{4}C + \frac{3}{4} + C = \frac{27}{12}$

$\boxed{C = \frac{2}{3}}$

$\boxed{A = \frac{5}{6}}$

$A - C \Rightarrow \frac{5}{6} - \frac{2}{3}$

$\Rightarrow \frac{1}{6}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Express $\frac{2}{3}$ of $\frac{1}{4}$ of Rs 25.20 as a fraction of $1\frac{1}{2}$ of Rs 36

Ans $\frac{2}{3} \times \frac{1}{4} \times 25.20 = x \times \frac{3}{2} \times 36$

$$\boxed{x = \frac{7}{90}}$$

Q ⇒ $8 - 8 \times \dfrac{2\frac{1}{5} - 1\frac{2}{7}}{2 - \frac{1}{6 - \frac{1}{6}}}$ equal to?

Ans ⇒ $8 - 8 \times \dfrac{\frac{11}{5} - \frac{9}{7}}{2 - \frac{1}{\frac{35}{6}}}$

$$\boxed{\Rightarrow 4}$$

Q ⇒ $8\frac{1}{6} + 5\frac{1}{8} + 4\frac{2}{3}$?

Ans ∵ $8 + 5 + 4 \Rightarrow 17$

$\frac{1}{6} + \frac{1}{8} + \frac{2}{3} \Rightarrow \frac{23}{24}$

$\Rightarrow 17\frac{23}{24}$

POORNIMA

RATIO & PROPORTIONS

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ $a : b = c : d$

$\frac{a}{b} = \frac{c}{d}$ } → Called Proportions

★ $a : b :: c : d$

$a : b$ as $c : d$

★ $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$

★ $\frac{a}{b} \not\rightarrow \frac{c}{d}$

$ad > bc \rightarrow \frac{a}{b} > \frac{c}{d}$

$ad < bc \rightarrow \frac{a}{b} < \frac{c}{d}$

$ad = bc \rightarrow \frac{a}{b} = \frac{c}{d}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

✡ $\frac{a}{b} = \frac{c}{d}$

$\Rightarrow \boxed{\frac{a+b}{b} = \frac{c+d}{d}}$ → Componendo Rule

✡ $\frac{a}{b} = \frac{c}{d}$

$\Rightarrow \boxed{\frac{a-b}{b} = \frac{c-d}{d}}$ → Dividendo Rule

✡ $\frac{a}{b} = \frac{c}{d}$

$\Rightarrow \boxed{\frac{a+b}{a-b} = \frac{c+d}{c-d}}$ → Componendo - Dividendo Rule

✡ $\frac{a}{b} = \frac{c}{d}$

$\Rightarrow \boxed{\frac{b}{a} = \frac{d}{c}}$ → Invertendo

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ $a : b : c$

- a, c are extreme proportions
- b is mean proportional

Q ⇒ Greater ratio, $17:18$ & $10:11$

Ans $\frac{17}{18} \not\gtrless \frac{10}{11}$

$187 > 180$

⇒ $\boxed{\frac{17}{18}}$

Q ⇒ The 3rd proportional to 18 and 54 is?

Ans ⇒ $a : b : c$

⇒ $a : b :: b : c$

$\frac{a}{b} = \frac{b}{c}$

$b^2 = ac$

$(54)^2 = 18c$ $\boxed{c = 162}$

$18 : 54 : 162$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ 4 proportional in 14, 13, 153

Ans ⇒ a : b : c : d

$$\frac{a}{b} = \frac{c}{d}$$

$$d = \frac{c}{a} \cdot b$$

$$d = \frac{153}{14} \times 13$$

$$\boxed{d = 221}$$

Q ⇒ Mean proportional b/w 7 & 63

Ans ⇒ a : b : c
7 ↓ 63
?

$$\Rightarrow \frac{a}{b} = \frac{b}{c}$$

$$\Rightarrow \frac{7}{b} = \frac{b}{63}$$

$$\Rightarrow b^2 = 441$$

$$\Rightarrow \boxed{b = 21}$$

OR

$$b^2 = 7 \times 63$$
$$b^2 = 7 \times 7 \times 9$$
$$b = \sqrt{49 \times 9}$$
$$= 7 \times 3$$

7 : 21 : 63

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ $\frac{10}{13} = \frac{11}{28} = \frac{21}{11} = \frac{12}{11} = K$. What is K?

Ans ⇒ Add N SD

⇒ $\frac{10+11+21+12}{13+28+11+11}$

⇒ $\boxed{\frac{6}{7} = K}$

Q⇒ Income Ratio of Ramesh & Suresh is 5:6. Their Spending Ratio is 7:9. Ramesh saves Rs 4000 & Suresh saves Rs 3000. Income & Spending respectively of Ramesh & Suresh are?

Ans Income = Spend + Saving

Spend = Income - Saving

⇒ Income Ratio (R:S) = (5:6)

Not actual value

* Actual Income R = 5k

" " S = 6k

$\frac{7}{9} = \frac{\text{Spend R}}{\text{Spend S}} = \frac{5K-4000}{6K-3000}$

$\boxed{K = 5000}$

Actual Income R = 25000.

S = 30000 ✓

Spending R = 21000

S = 27000 ✓

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ a : b = 3 : 7 What is a : b : c ?

b : c = 9 : 5

Ans

3 × 9 : 7 × 9 9 × 7 : 5 × 7

27 : 63 63 : 35

27 : 63 : 35

Q ⇒ How to divide 3395 in Ratio of 42 : 32 : 23

Ans (I) Method ⇒

$\Rightarrow \frac{42}{42+32+23} \times 3395 \Rightarrow 1470$

$\Rightarrow \frac{32}{42+32+23} \times 3395 \Rightarrow 1120$

⇒ 3395 − (1470 + 1120)

⇒ 805

1470 : 1120 : 805

(II) Method ⇒

Let Actual value be 42K, 32K, 23K

42K + 32K + 23K = 3395

K = 35

42 (35) ⇒ 1470 — — —

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ 285 is summation of 3 numbers. Ratio b/w 2nd & 3rd numbers is 6:5. Ratio b/w 1st & 2nd number is 3:7. The 3rd number is?

Ans

a : b	b : c
3 : 7	6 : 5
3×6 : 7×6	6×7 : 5×7
18 : 42	42 : 35

⇒ 18 : 42 : 35

⇒ 3rd number ⇒ $\frac{35}{18+42+35} \times 285$

⇒ 105

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Ratio of 2 numbers is 3:8. On adding 5 to both numbers, the ratio become 2:5. Which is the smaller number out of two.

Ans ⇒ A:B (3:8)

⇒ Let actual no be 3K, 8K

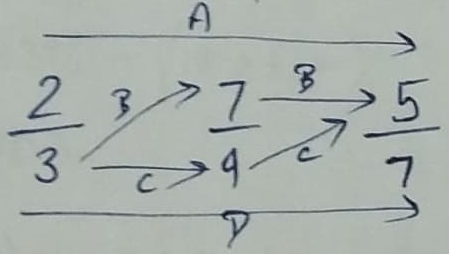
⇒ $\frac{3K+5}{8K+5} = \frac{2}{5}$

$\boxed{K=15}$

⇒ 3K ⇒ ? A is smallest
8K ⇒ ?

Q ⇒ Find A:B:C:D when A:B = 2:3, B:C = 7:9, C:D = 5:7

Ans



⇒ A = Multiply all N ⇒ 2×7×5 ⇒ 70

⇒ D = Multiply all D ⇒ 3×9×7 ⇒ 189

⇒ B = 3×7×5 ⇒ 105

⇒ C = 3×9×5 ⇒ 135

70 : 105 : 135 : 189

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Price of each Article of type P, Q, R is 300, 180, 120. Suresh buys articles of each type in ratio 3:2:3 in Rs 6480. How many articles of type Q did he purchase.

Ans ⇒

P	Q	R
Rs 300	Rs 180	Rs 120
↓	↓	↓
3k	2k	3k

$900k + 360k + 360k = 6480$

$\boxed{k = 4}$

$Q = 2k$

$\boxed{Q = 8}$ articles of Q

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Ajay & Raj together have Rs 1050. On taking Rs. 150 from Ajay, Ajay will have same amount as what Raj had earlier. Find the ratio of amounts with Ajay & Raj initially.

Ans ⇒ Let 'A' & 'R' be Initially

⇒ $A + R = 1050$ — (1)

$A - 150 = R$

⇒ $A - R = 150$ — (2)

⇒ $A + R + A - R = 150 + 1050$

$\boxed{A = 600}$

⇒ $A + R = 1050$

$\boxed{R = 450 Rs}$

⇒ $A : R = 600 : 450$

$\boxed{A : R = 4 : 3}$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ If $x:y = 3:4$, then $(7x+3y):(7x-3y)$ equals to ?

Ans (I) Method ⇒

$$\frac{x}{y} = \frac{3}{4} \Rightarrow \boxed{\begin{matrix} x=3 \\ y=4 \end{matrix}}$$

$$\frac{7x+3y}{7x-3y} = \frac{21+12}{21-12} = \frac{11}{3}$$

(II) Method ⇒

$$\Rightarrow \frac{x}{y} = \frac{3}{4} \quad [\text{Multiply N by 7 \& D by 3}]$$

$$\Rightarrow \frac{7x}{3y} = \frac{21}{12}$$

$$\Rightarrow \frac{7x+3y}{7x-3y} = \frac{21+12}{21-12} \quad \left[\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}\right]$$

$$= \frac{11}{3}$$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ If $a:b = 5:7$ & $c:d = 2a:3b$, then $ac:bd$ is

Ans $\frac{a}{b} = \frac{5}{7}$ | $\frac{c}{d} = \frac{2a}{3b}$

$\Rightarrow \frac{a}{b} \times \frac{c}{d} = \frac{5}{7} \times \frac{2a}{3b}$

$\Rightarrow \frac{ac}{bd} = \frac{10a}{21b}$ $\left[\begin{matrix} a = 5 \\ b = 7 \end{matrix}\right]$

$\Rightarrow ac:bd \Rightarrow 50:147$

Q ⇒ 3 no are in ratio $\frac{1}{2}:\frac{2}{3}:\frac{3}{4}$. The difference b/w greatest & smallest no is 36. Find numbers.

Ans $\Rightarrow \frac{1}{2}k : \frac{2}{3}k : \frac{3}{4}k$

$\Rightarrow \underbrace{0.5k}_{\text{Small}} : 0.606k : \underbrace{0.75k}_{\text{greater}}$

$\Rightarrow \frac{3}{4}k - \frac{k}{2} = 36$

$\Rightarrow \boxed{k = 144}$

$\Rightarrow$ 72
96
108

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Ravi & Karan salary in 2:3. If each salary increased by 5,000, ratio become 40:55. New salary of Karan.

Ans Trick ⇒

$$\text{dif} = 38 \leftarrow \begin{matrix} 2 : 3 \\ 40 : 55 \end{matrix} \rightarrow \text{diff} = 52$$

⇒ Since diff is not same, multiply difference of ratio with other

⇒ $\overset{1}{\frown}$ 2 : 3

40 : 55 $\underset{15}{\smile}$

⇒ 30 : 45 (10)

(10) 40 : 55

⇒ $\frac{5000}{10} \Rightarrow \underline{500}$

⇒ Karan's Salary ⇒ 500 × 55

⇒ 27,500

Ravi " ⇒ 500 × 40

⇒ 20,000

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

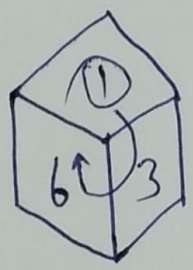
DICE

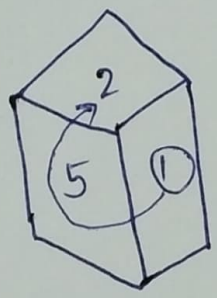
Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ 2 types ⇒ Standard (Sum of opp face is always 7)
⇒ Non-standard (General)

★ If sum of 2 adjacent faces is 7, then non-standard dice.

Q ⇒





opp of 6?

Ans Pick common no & Rotate clock or Anti-clockwise

1	3	6
	↕	↕
1	5	2

(4)

opp of 6 is 2

★★ Non standard dice as adjacent sum is "7"

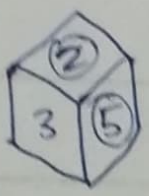
★★ If dice is standard then, 6 + ? = 7

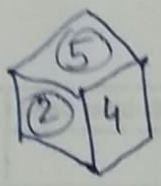
(? = 1)

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q.)





opp of 3?

Ans ✱ Since 2 & 5 are common in both dices then,

3 is opp of 4

✱ Also we can do this

2	5	3
2	5	4





POORNIMA

Height & Distance

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★★ $\sqrt{3} = 1.73$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

★ $^{8}C_{5} = \frac{8!}{5!\,3!} = 56$

OR

★★ Easy & fast

$$\frac{5 \text{ times (Back)}}{5 \text{ times (Start)}} \Rightarrow \frac{8 \times 7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4 \times 5} = 56$$

POORNIMA

PERMUTATION & COMBINATION

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Arrange 7 items in a line ⇒ $(7!)$

★ Arrange 7 items in a circular ⇒ $(7-1)!$

⇒ $(6)!$

★ Permutation ⇒ Select & arrange

Combination ⇒ Select only

★ Combination ⇒

$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

★ Permutation ⇒

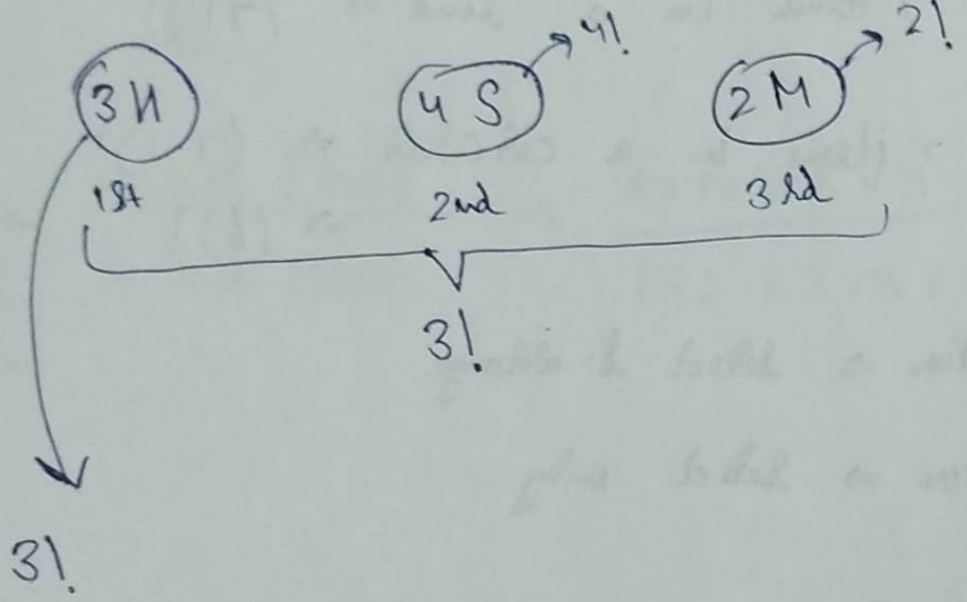
$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ 3 History, 4 Science, 2 Maths. In how many ways arrange the books so that all books of same subject are together.

Ans



⇒ 3! × 3! × 4! × 2!

⇒ 1728 ways.

Q⇒ Arrange "FUZZTONE" so all vowels together.

Ans Vowel ⇒ (UOE) → 3!
↳ 1st group

Total
↳ Vowels + others ⇒ 1 + 5
⇒ (6!)

$$\frac{3! \times 6!}{(2!)} \Rightarrow \underline{2160}$$

↳ Z is 2 times.

3! (n-2)!

$$\Rightarrow \frac{3! \times 6!}{2!}$$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ Arrange "MARKER", such that no vowels together

Ans ⇒ Total ways - Vowel together

⇒ $\frac{6!}{2!} - \frac{2! \times 5!}{2!}$

Vowel group + other than vowels

240 ways

Q⇒ Without Repetition, using 2, 3, 4, 5, 6, 8, 0. How many no made b/w 500 & 1000?

Ans $\underline{3} \times \underline{6} \times \underline{5} \Rightarrow 90$

OR

$^{3}C_{1} \times {}^{6}C_{1} \times {}^{5}C_{1} \Rightarrow 90$

Out of 5, 6, 8. Select only 1

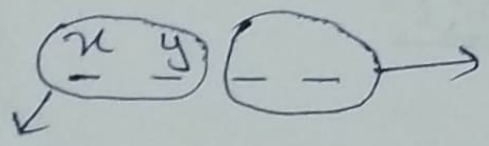
Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ 4 members from a group out of 8 members.

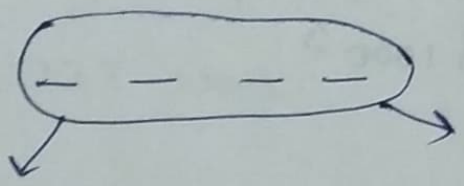
(i) In how many ways it is possible to make the group if 2 particular members must be included.

(ii) In how many ways it is possible to make the group if 2 particular members must not be included.

Ans (i) 

$^6C_2 \Rightarrow \frac{6!}{2!\,4!}$

$\Rightarrow 15$

(ii) 

x & y not be included

$^6C_4 \Rightarrow \frac{6!}{4!\,2!}$

$\Rightarrow 15$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ There are 8 routes London to Delhi & 6 routes delhi to Tokyo. In how many different way Raj can travel from London to Tokyo via Delhi.

Ans London to delhi ⇒ $^8C_1$

Delhi to Tokyo ⇒ $^6C_1$

$^8C_1 \times {}^6C_1 \Rightarrow \underline{48}$

Q ⇒ There are 35 people. 12 girls, 10 boys, 5 senior, 8 lady. Select a leader from girl or boy. In how many way can be done.

Ans $12 + 10 \Rightarrow 22$

↓ ↓

Girl Boys

$\Rightarrow {}^{22}C_1 \Rightarrow \frac{22!}{1!\,21!} \Rightarrow \underline{22}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ In a class, there are 15 students. During a christmas party all of them shook hands with each other only once. How many handshake took place.

Ans ⇒ 2 person needed for handshake.

$$\Rightarrow {}^{15}C_2 \Rightarrow \frac{15!}{2!\,13!} \Rightarrow 105$$

Q ⇒ A bank has 6 digit account no. with no repetation. The first & last digit are fixed. How many account can be possible?

Ans (x) 8 × 7 × 6 × 5 (y)

⇒ 1680

x & y are fined

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ A trekking group is to be formed having 6 members. They are to be selected from 3 G, 4 B, 5 T. In how many ways can the group be formed so that 3T and 3B or 2G and 4T

Ans

AND = *
OR = +

** Dependent event

Independent event

(3T and 3B) or (2G and 4T)

$$\left({}^{5}C_{3} * {}^{4}C_{3}\right) + \left({}^{3}C_{2} * {}^{5}C_{4}\right)$$

⇒ 55

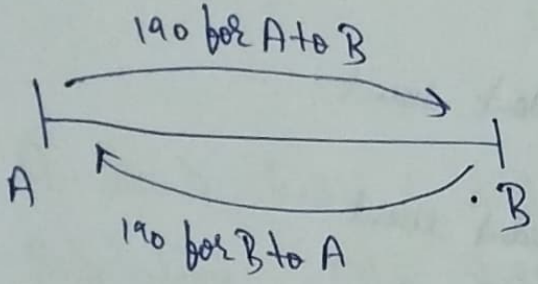
Main Ideas, Questions & Summary:

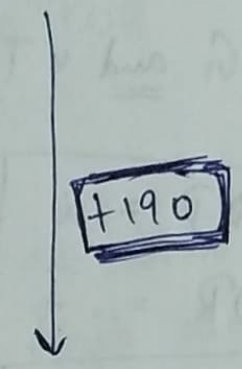
Library / Website Ref.:-

Q ⇒ On a railway line there are 20 stops. A ticket is needed to travel b/w any 2 stops. How many different tickets would the gov. need to prepare to cater to all possibilities.

Ans

$^{20}C_2 \Rightarrow \frac{20!}{2!\,18!} \Rightarrow 190$

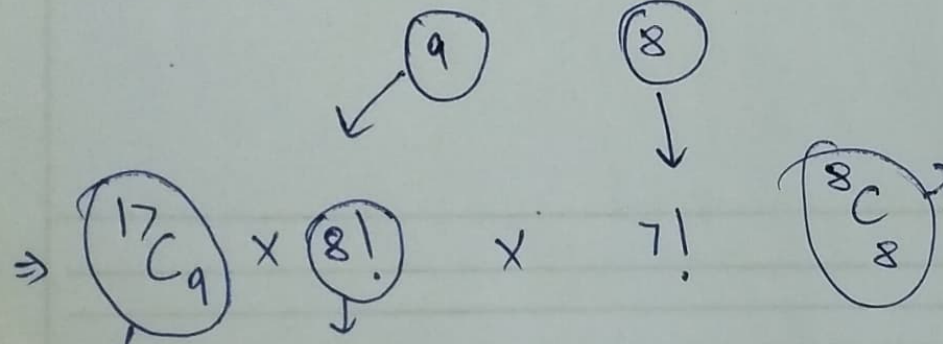




380

Q ⇒ 17 students are present in a class. In how many ways, can they be made to stand in 2 circles of 8 and 9 students?

Ans ⇒ for circle $(n-1)!$



POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ In a room there are 2 G, 3 Y, 4 B chairs. In how many ways can Raj choose 3 chairs so that at least one Y chair is included?

Ans ⇒ Total way = No Y + Choosing Y

~~No~~ ⇒ Choose Y = Total Way − No Y

⇒ $^{9}C_{3} - {}^{6}C_{3}$

⇒ 20 ways.

Q ⇒ In how many ways a word of 5 different letters can be formed using 2 vowel at a time.

Ans $^{21}C_{3} \times {}^{5}C_{2} \times (5!)$

Select the letters

Used for different arrangement.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

SIMPLE INTEREST

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

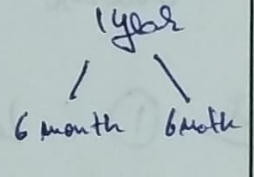
★ $SI = \frac{PTR}{100}$

P = principle
T = Time
R = Rate of interest

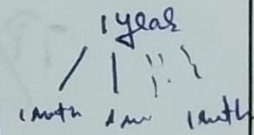
★ Total Amount after SI ⇒

$$\text{Amount} = P + SI$$

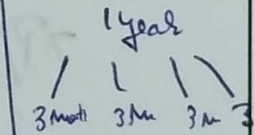
★ If T = half yearly ⇒ $SI = \frac{P(R/2)(2T)}{100}$



★ If T = 1 month ⇒ $SI = \frac{P(R/12)(12T)}{100}$



★ If T = quaterly ⇒ $SI = \frac{P(R/4)(4T)}{100}$



Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Amount of 6500. at quarterly interest of 8%, will yield how much in 2 & half years.

Ans $SI = \frac{PRT}{100} \Rightarrow \frac{6500\,(8/4)(4\times 2.5)}{100} \Rightarrow 1300$

$$\text{Total} = 6500 + 1300 = 7800$$

Q ⇒ Amount becomes 7 times in 15 yrs. In how many yrs will the same amount become 10 times? The rate of interest remains same for both cases.

Ans ① Let amount be 'P'.

$$7P = \frac{PRT}{100}$$

$$7P = \frac{P\times R\times 15}{100}$$

$$7P = P + \frac{PRT}{100}$$

$$6P = \frac{P\times R\times 15}{100}$$

$$\boxed{R = 40\%}$$

②

$$10P = P + \frac{P\times 40\times T}{100}$$

$$\boxed{T = 22.5 \text{ yrs}}$$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Sum becomes 3000 at rate of 12% per annum. The same sum become 3300 at rate of 15% per annum in same duration. Find sum & duration.

Ans (3000) $Amt_1 = P + SI_1$ — (1)

(3300) $Amt_2 = P + SI_2$ — (2)

$\Rightarrow Amt_2 - Amt_1 = 300$

$\Rightarrow (P + SI_2) - (P + SI_1) = 300$

$\Rightarrow \frac{P \times 15 \times T}{100} - \frac{P \times 12 \times T}{100} = 300$

$\Rightarrow \boxed{PT = 10,000}$ — (3)

In (1),

$SI_1 = \frac{PTR}{100}$

$SI_1 = \frac{10,000 \times 12}{100}$

$\boxed{SI_1 = 1200}$

$3000 = P + 1200$

$\boxed{P = 1800}$

↳ Using (3)

$PT = 10,000$

$\boxed{T = \frac{50}{9} \text{ yrs}}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ A man got 130 less, as SI, when he invested 2000 for 4 yrs as compareded to investing 2250 for same duration. What is Rate of interest.

Ans ⇒ $SI_2 = SI_1 + 130$

⇒ $SI_2 - SI_1 = 130$

⇒ $\frac{2250 \times R \times 4}{100} - \frac{2000 \times R \times 4}{100} = 130$

⇒ $\boxed{R = 13\cdot1\%}$

Q ⇒ 1/5 part amount given at 3%, 1/3 at 5%, 2/5 at 9%, Remaining at 11%. Total interest Recieved 297. P?

Ans $SI = SI_1 + SI_2 + SI_3 + SI_4$

$$\Rightarrow 297 = \frac{(P/5) \times 3 \times 1}{100} + \frac{(P/3) \times 5 \times 1}{100} + \frac{(2P/5) \times 9 \times 1}{100} + \frac{\left(P - \frac{P}{5} - \frac{P}{3} - \frac{2P}{5}\right) \times 11}{100}$$

⇒ $\boxed{P = 4500}$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ 20,400 was divided in 2 parts & then invested. One part invested at 6.25% for 8 yrs yield same interest as other part invested at 7% for 5 yrs. What is value of smaller part.

Ans

20,400

(x) (20,400 − x)

$\Rightarrow SI_1 = SI_2$

$\Rightarrow \frac{x \times 6.25 \times 8}{100} = \frac{(20,400 - x) \times 7 \times 5}{100}$

$\Rightarrow \boxed{x = 8400 \text{ Rs}}$ → Smaller part

$\Rightarrow 20,400 - x$

20,400 − 8400 → Larger part

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ If SI on certain sum of 15 months at 7(1/2)% p.a exceeds SI on same sum for 8 months at 12(1/2)% p.a by 32.50, then sum is

Ans ⇒ $SI_1 - SI_2 = 32.50$

$$\Rightarrow \frac{P \times 15 \times 7.5}{100 \times 12} - \frac{P \times 12.5 \times 8}{100 \times 12} = 32.50$$

Converted to year

$$\Rightarrow \boxed{P = 3120}$$

Q ⇒ If SI for 2 yrs. for a sum is 600 & C.I for same sum for 2 yrs & same rate of intrest is 645, what will be rate of intrest.

Ans

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Aman invests 8,000 at some rate of interest. Being SI the money doubles in 5yrs. Raj see this & invests 6250 for 3yrs at same rate of interest. How much interest does Raj get?

Ans

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

PERCENTAGE

$\frac{1}{7} = 14.28\%$ | $\frac{2}{7} = 28.57\%$

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ If price of rice is 30% less than that of wheat, then price of wheat is how much per cent more than that of rice?

Ans ⇒ Let Wheat = 100

Rice = 70

⇒ $\frac{100}{70} \times 100$ $\left[\because \frac{A}{B} \times 100\right]$

⇒ 142.85 %.

⇒ 142.85 − 100

⇒ 42.85 %.

⇒ $\frac{30}{100-30} \times 100$

⇒ 42.86 %.

$\frac{R}{100 \pm R} \times 100$

Q ⇒ Price of A is first inc by 10%. & then dec by 10%. What is change in price of apple.

Ans Let price be 100

⇒ $+10 - 10 + \left(\frac{-10 \times 10}{100}\right)$

⇒ −1 %.

⇒ $100 - \left(100 \times \frac{1}{100}\right)$

⇒ 99 → New price

Change is of 1 Rs

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Y has to score 40% marks to pass. He gets 20 marks & fails by 40 marks. The max marks of examare?

Ans

40% → 60

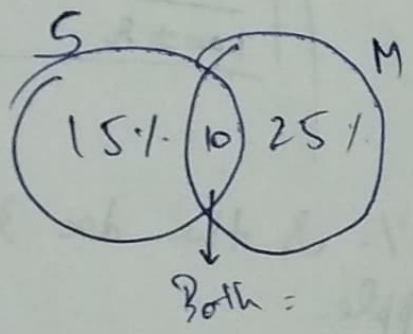
1% → 1.5

So 100% ⇒ 1 × 1.5

⇒ 150

Q ⇒ In a class, 15% fail in S, 25% fail in M, 10% fail in both (S & M). What is % of pass.

Ans



Fail ⇒ 15 + 25 − 10

⇒ 30%

Pass ⇒ 70%

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ By 20% decrease in price of rice, people can buy 10kg more rice in Rs100. What is original price of 1kg of rice?

Ans

100 Rs → 10kg

$\frac{100}{10}$ = 1kg

10 Rs → 1kg

⇒ $\frac{20}{100}$ ⇒ $\frac{1}{5}$

⇒ 5 → ?

4 → 10

⇒ $\frac{10}{4}$ = 2.5 Rs

No need to multiply by 5 as it will not give value of 1 kg.

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ In election, one candidate got 40% of total votes and yet lost by 1000 votes. What is total no. of votes casted in election?

Ans

$$\frac{40}{100}x + 100 = \frac{60}{100}x$$

$\boxed{x = 5000}$ → Total votes.

Q ⇒ In country 55% is female. 80% of male is literate. How much of female are literate if total literacy is 58%?

Ans ⇒ 55% Female Let Total Pop = 100

45% male

⇒ $\frac{80}{100} \times 45$ ⇒ 36 male literate

⇒ Total literate ⇒ 58

M + F = Total

36 + F = 58

$\boxed{F = 22}$ → It's our assumption out of 100 pop

∴ original % ⇒

$$\frac{22}{55} \times 100$$

⇒ 40%

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q⇒ If 20% of bill is deduced, then 100 is still to be paid. How much was orignal bill?

Ans $x - \frac{20}{100}x = 100$

$\boxed{x = 125}$

Not confirm correct or not.

$5 \rightarrow ?$

$4 \rightarrow \frac{100}{4} \times 5$

Q⇒ Two numbers are less than a third no by 30% & 37%. How much % is second no less than the first?

Ans ⇒ $A + \frac{30}{100}C = C$

$\boxed{\frac{A}{C} = \frac{70}{100}}$

⇒ $B + \frac{37}{100}C = C$

$\boxed{\frac{B}{C} = \frac{63}{100}}$

⇒ A = 70

B = 63

(70 - 63) ↳ less que asked

⇒ $\frac{7}{70} \times 100$ ⇒ 10%

OR

C = 100

$A = 100 - \frac{30}{100} \times 100$ ⇒ 70

$B = 100 - \frac{37}{100} \times 100$ ⇒ 63

⇒ $\frac{7}{70} \times 100$

⇒ 10%

Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

ELEMENTARY STATISTICS

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Mean ⇒ $\frac{\text{Sum of all observation}}{\text{No of observation}}$

- Min val of observation < Mean < Max val of observation

★ Mod ⇒ Highest freq of observation

⇒ Mode > 1 (freq)

★ Median ⇒ Central item

⇒ First arrange all observation in asc/des order.

$n \rightarrow$ odd $\Rightarrow \left(\frac{n+1}{2}\right)$ term

$n \rightarrow$ even $\Rightarrow \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2}+1\right) \text{ term}}{2}$

★ Range = Max Val - Min Val

★ 3 Median = Mode + 2 Mean → Emperical formula

★ 3, 3, 4, 4

Mod ⇒ 3, 4 (Bimodal data leads)

★ 1, 2, 3, 4

No mod

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆☆ It is compulsory to have mean, median of dataset but not necessary to have mode.

Eg ⇒ 1, 2, 3, 4

☆☆ There should be only one mean, median of data but can have multiple mode.

☆ Standard deviation (σ) & Variance (σ^2) ⇒

$$\Rightarrow \sigma = \sqrt{\frac{\Sigma(x_i - \text{mean})^2}{N}}$$

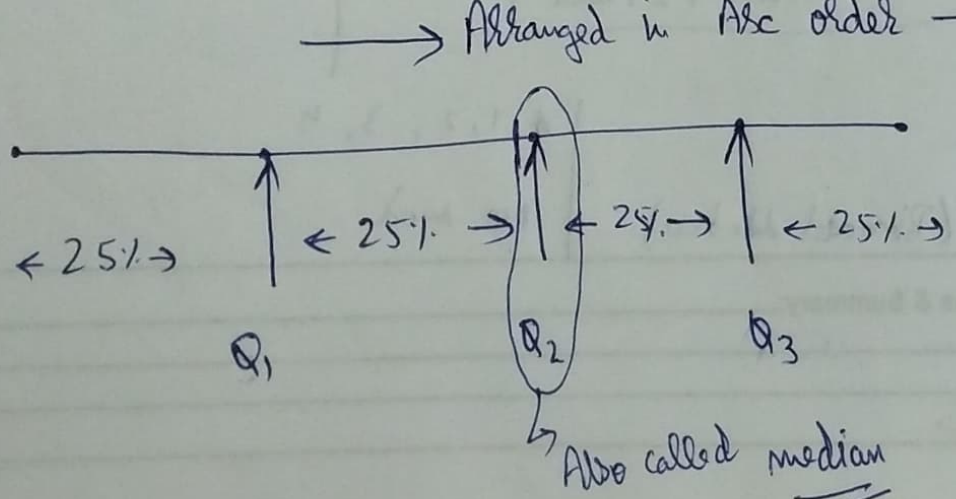
$\sigma = \sqrt{\text{variance}}$

$\sigma^2 = \text{variance}$

☆ Co-efficient of variance ⇒ Answer in %.

$$\Rightarrow \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

☆ Quartile ⇒



POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ 1st $(Q_1) = \frac{n+1}{4}$ term

$Q_2 = \frac{2(n+1)}{4}$ term

$Q_3 = \frac{3(n+1)}{4}$ term

★ Interquartile Range (IQR) ⇒ $Q_3 - Q_1$

Quartile Deviation ⇒ $(Q_3 - Q_1)/2$

★ Coff of Quartile Devi ⇒

$$\left(\frac{Q_3 - Q_1}{Q_3 + Q_1}\right)$$

★ Mean deviation ⇒

$$\frac{\Sigma |x_i - \text{mean}|}{N}$$

★ Mean Deviation = $\frac{4}{5}$ × Quartile deviation

★ Standard deviation (σ) = $\frac{5}{4}$ × Mean deviation

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ For table like 10-20, 20-40 ...

~~Standard~~

$$\text{Variance}(\sigma^2) = \frac{\sum f_i (x_i - \text{mean})^2}{\sum f_i}$$

POORNIMA

NUMBER SYSTEM

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ Factors ⇒ all no' that can completely divide a specific no'.

⇒ All no' (factors) are smaller/= than specific no'.

Eg ⇒ $12 = \{1, 2, 4, 6, 12\}$

★ $a \times b = c$;

then a & b are factors of c

$\frac{c}{a} = b \mid \frac{c}{b} = a$

★ Finding factors ⇒

① Prime factorization ⇒

2	12
2	6
3	3
	1

Should see prime

⇒ $12 = 2^2 \times 3^1$

★ No of factors = $(2+1)(1+1)$

$= 6$

② Trick Start ⇒ $12 \rightarrow 4 \times 3$ (Take any no'. Eg ⇒ 2×6)

$2 \times 2 \times 3$

Main Ideas, Questions & Summary:

↳ $2^2 \times 3^1$

↳ $(2+1)(1+1)$

⇒ 6

Library / Website Ref.:-

✱ Products of factors ⇒

$$(x)^{\frac{M}{2}} \rightarrow \text{No. of factors}$$

⇒ P_n ⇒ 12 ⇒ M = 6

$$(12)^{\frac{6}{2}} \Rightarrow (12)^3$$

✱ Magic of 9 ⇒

$9^2 = 81$

$99^2 = 9801$

$999^2 = 998001$

$99999^2 = \underline{9999}\ \underline{8}\ \underline{0000}\ \underline{1}$

$9^3 = 729$

$99^3 = 9\cdot 7\ 0\ 2\ 99$

$999^3 = 99\ 7\ 00\ 2\ 999$

$9999^3 = \underline{999}\ \underline{7}\ \underline{000}\ \underline{2}\ \underline{9999}$

✱ Magic of 11 ⇒

$11^2 = 121$

$111^2 = 12321$

$1111^2 = \underline{1234}\ \underline{321}$

✱ Remainders ⇒

$$q \overline{)\ p\ }\ r$$

$$s$$

p ⇒ dividend

q ⇒ divisor

r ⇒ quotient

s ⇒ remainder

$$\boxed{p = qr + s}$$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ No divided by 406 leaves remainder 115, what will be remainder when it will be divided by 29?

Ans Trick ⇒ $\frac{\text{Divisor} + \text{Given Rem}}{\text{Target Divisor}} \Rightarrow \frac{406 + 115}{29} = 28$

OR

$\frac{\text{Given Rem}}{\text{Target Divisor}} = \frac{115}{29} = 28$ → ★ Ans is remainder not quotient ★

↳ $\frac{406}{29} + \frac{115}{29}$

↳ Given remainder 0 always.

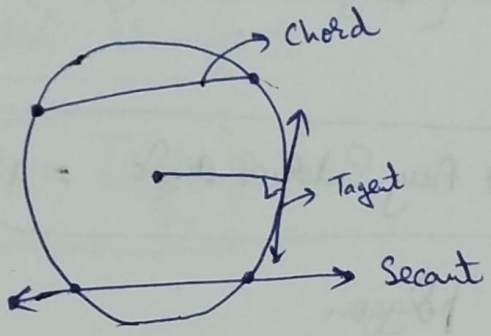
Main Ideas, Questions & Summary:

Library / Website Ref.:-

POORNIMA

MENSURATION

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-



★ Regular polygon ⇒ all sides & angle are equal of a polygon. (Interior & exterior)

★ Interior angle sum ⇒ $(n-2) \times 180^\circ$

- Triangle (3) ⇒ $(3-2) \times 180$
  ⇒ 180°

★ Individual angle ⇒

$$\frac{(n-2) \times 180}{n}$$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

☆ Interior angle + Corresponding Exterior angle = 180°

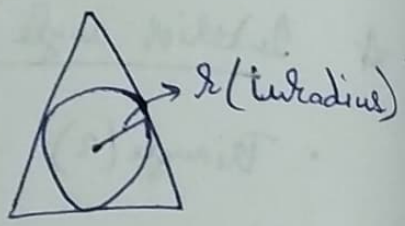
↳ For all polygon (regular & irregular)

☆ Any Interior angle + Any Exterior angle = 180°

↳ Only for regular polygon

Q ⇒ What is measure of radius of circle inscribed in a triangle whose sides measure 8, 15, 17 units ?

Ans ☆ $r = \frac{\text{Area of } \Delta}{\text{Semi perimeter } \Delta}$



☆ Semi-perimeter (s) $= \frac{a+b+c}{2}$

☆ Ar of $\Delta \Rightarrow \sqrt{s(s-a)(s-b)(s-c)}$

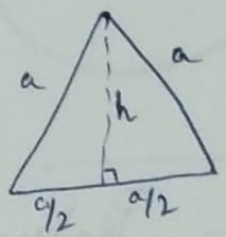
☆ If $\angle$ triangle ⇒

Short trick ⇒ $r = \frac{b+p-h}{2}$

POORNIMA

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

★ If equilateral triangle ⇒



• By fig ⇒ $\boxed{h = \frac{\sqrt{3}\,a}{2}}$

• $r = \frac{\text{Area } \Delta}{\text{Semi perimeter } \Delta}$ ⇒ $\frac{\frac{1}{2} \times a \times \frac{\sqrt{3}a}{2}}{\frac{a+a+a}{2}}$

Short trick $\boxed{r = \frac{a}{2\sqrt{3}}}$ or $\boxed{r = \frac{h}{3}}$ → If h is given only ↓ height

Ans So, in que, given Δ is ∠ $(17)^2 = (8)^2 + (15)^2$

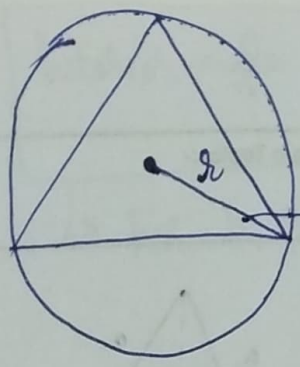
$r = \frac{8+15-17}{2}$

$\boxed{r = 3}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Circum Circle



→ Circum Radius

⇒ $$R = \frac{\text{Product of sides of } \Delta}{4 * \text{Area of } \Delta}$$

★ If $\angle$ triangle ⇒

$$R = \frac{\text{hypothenus}}{2}$$

★ Equilateral Δ ⇒

$$R = \frac{a}{\sqrt{3}}$$ or $$R = \frac{2}{3}h$$ → Height is given

POORNIMA

PROFIT & LOSS

Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Ajay incurred loss of 20% by selling a vase for 2880. To get profit of 20% a what price should he sell the vase?

Ans $\boxed{SP = \frac{100 \pm x}{100} \times CP}$

$\Rightarrow 2880 = \frac{100 - 20}{100} \times CP$

$\boxed{CP = 3600}$

$\Rightarrow \begin{array}{r} 3600 \\ 2880 \\ \hline 720 \end{array} \rightarrow \begin{array}{r} 3600 \\ 720 \\ \hline 4320 \end{array}$ =

OR

$SP = \frac{100 + 20}{100} \times 3600$

$\boxed{SP = 4320}$

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q ⇒ Ram sells planes at rate of 20 planes for Rs1. If he gets profit of 20%. How many planes did he buy in 1 Rs.

Ans

$$1 = \left(\frac{100+20}{100}\right) \times CP$$

$$\boxed{CP = \frac{5}{6}} \rightarrow \text{if } 20 \text{ planes} \rightarrow \frac{5}{6} \text{ Rs}$$

$$? \rightarrow 1 \text{ Rs}$$

$$20 = \frac{5}{6} \times ?$$

$$? = \frac{120}{5} \Rightarrow \underline{24 \text{ planes}}$$

Q ⇒ Uma wants to gain 15% profit on sugar. She buys 120 kg at ₹ 24/kg to mix with 180 kg of sugar bought at 28/kg. She sells sugar mix at?

Ans

$$120 \times 24 \Rightarrow 2880$$

$$180 \times 28 \Rightarrow \underline{5040}$$

$$\underline{7920}$$

$$SP = \frac{115}{100} \times 7920$$

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Date	Unit No.	Lecture No.	Faculty	Subject Name	Subject Code	Main Topics:-

Q ⇒ Ram buys some oranges in a shop at 4 per rupee. He goes to other shop & buys same no of orange at 5 per rupee. He then combines them in a basket & sell them at 4 per rupee. Will she get a profit or loss? And how much?

Let 20 at Shop1
20 at Shop2

Total Oranges ⇒ 40

Ans ⇒ 4 Orange → 1 Rs
20 Orange → ?
5 Rs

⇒ 5 Orange → 1 Rs
20 Orange → ?
4 Rs

~~⇒ 40 Oranges → 4 Rs~~

⇒ 4 Oranges → 1 Rs
40 Oranges → ?
10 Rs

⇒ P/L = SP − CP
= 10 − 9
= 1 Rs profit → $\frac{1}{9} \times 100$ ⇒ 11.11% profit

* Trick ⇒

⇒ $\frac{2 \times R_1 \times R_2}{R_1 + R_2}$ → Mean Cost per Unit

⇒ $\frac{2 \times \frac{1}{4} \times \frac{1}{5}}{\frac{1}{4} + \frac{1}{5}}$

⇒ $\frac{2}{9}$ per Orange

⇒ (SP − CP)/CP × 100

⇒ $\frac{\frac{1}{4} - \frac{2}{9}}{\frac{2}{9}} \times 100$ ⇒ 11.11%

Main Ideas, Questions & Summary:

Library / Website Ref.:-

Q⇒ Ram bought food worth 56,000. He then sold 1/3 rd of it at loss of 40%. What profit he must earn on rest of food to nullify this loss?

Ans

$$\text{Profit\% needed} = \frac{\text{Loss\%} \times \text{Sold fraction}}{\text{Remaining fraction}}$$

$$\Rightarrow \frac{40 \times 1}{2} = 20\%$$